Oil-at-Risk:

A Framework for Studying Oil Price Vulnerabilities

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Abstract

Crude oil prices act as a key macroeconomic variable, often serving as a primary input in global inflation dynamics, industrial production costs, and international trade balances. The propagation of geopolitical tensions through energy markets can generate heightened volatility, which carries macroeconomic consequences and potential systemic risks for policymakers, firms, and institutional investors. Many traditional econometric approaches assume that the average historical effect of a geopolitical shock adequately represents the underlying stochastic process. However, in the context of crude oil prices and geopolitical shocks, where data tend to exhibit substantial heteroskedasticity, asymmetric responses, and influential tail observations that can distort linear estimates, this assumption may prove overly restrictive. To address the limitations of conditional mean forecasting, the present research adapts the *Growth-at-Risk* framework to model the distribution of future crude oil price growth. Rather than focusing solely on average baseline effects, the analysis estimates how fluctuations in geopolitical risk shift specific quantiles of the return distribution. This approach aims to capture heterogeneity across outcomes and to describe both downside and upside market vulnerabilities. The empirical strategy relies on the European Brent crude oil spot price and a dictionary-based Geopolitical Risk Index. Drawing on daily returns and a set of control variables, including a realised-volatility proxy, the U.S. dollar index, and a shipping-cost measure, the study specifies multiple quantile regressions to isolate the tail-risk imprint of geopolitical uncertainty at a one-trading-week horizon. A backtesting framework combining Kupiec, Christoffersen, and dynamic quantile tests evaluates the conditional risk measures against realised outcomes, and an entropy-based indicator is examined for early-warning content. The full analytical pipeline is implemented in a self-contained, open-source Python module to promote replicability.

KEYWORDS: Geopolitical Risk, Brent Crude Oil, Quantile Regression, Growth-at-Risk, Tail Risk, Volatility, Johnson S_U, CAViaR, Expected Shortfall, Kullback–Leibler Divergence.

JEL CODES: C21, C22, C53, C58, F51, G17, Q43.

1 Introduction

The study of geopolitical risk and its cascading effects on crude oil prices remains a significant concern for the global economy. While inflation dynamics vary due to a multitude of complex macroeconomic factors, oil prices serve as one of their primary drivers, often influencing the cost of global industrial production and affecting international trade balances and current account deficits. Geopolitical shocks, ranging from localised military conflicts and diplomatic standoffs to sweeping international embargoes and maritime blockades, tend to propagate rapidly through energy markets. These events can introduce profound uncertainty that may prompt immediate adjustments in consumption, investment, and precautionary storage among economic agents. Understanding not only whether geopolitical risk affects oil prices, but how vulnerable future prices might be to the extreme tails of these global risks, is therefore a valuable endeavour for economic stabilisation and financial risk management.

Standard macroeconomic and econometric frameworks have historically tended to favour modelling the conditional mean of asset prices and macroeconomic indicators. By frequently relying on Ordinary Least Squares (OLS) regression techniques, classical studies often estimate the average historical effect of a geopolitical shock, treating variance as a relatively constant dispersion around that mean. However, the observable dynamics of global oil markets frequently challenge this setup. Energy market data commonly exhibit intense volatility clustering, asymmetric shock responses, and heavy-tailed distributions. In this case, the average effect may be skewed by extremes without adequately capturing them, which can leave policymakers and risk managers with a misleading sense of both the magnitude and the likelihood of adverse price movements.

This empirical challenge aligns with an extensive literature in energy econometrics regarding the potential non-linearities of the oil market. As argued by Hamilton (2003), modelling oil shocks often benefits from a flexible approach to account for the possibility that price increases may have a different macroeconomic impact than price decreases. Kilian and Vigfusson (2011) provide evidence suggesting that the macroeconomic response to energy shocks can be asymmetric. Consequently, a strictly linear methodology risks understating downside vulnerabilities. The exposure of the market may not be adequately captured by mean forecasts; it typically requires a framework specifically equipped to describe asymmetric impacts.

Geopolitical shocks introduce an additional layer of non-linear complexity. Pinchetti (2024) suggests that large geopolitical shocks raise Brent prices and inflation expectations by triggering precautionary wait-and-see behaviours and a surge in speculative demand for crude. This is further contextualised by Blanchard and Galí (2010), who highlight how modern oil shocks interact differently with the macroeconomy than they did in the 1970s, mediated by factors such as wage rigidities, central bank credibility, and the shifting energy intensity of the economy. The physical materialisation of a threat is not required to damage the economy: Kilian et al. (2026) find that an unanticipated increase in the probability of a major geopolitical oil production shortfall, a perceived disaster scenario, can be sufficient to generate a marked surge in oil prices and macroeconomic uncertainty. The *Growth-at-Risk* methodology, adapted here as *Oil-at-Risk*, offers an econometric solution suited to this dynamic. By isolating specific quantiles, it describes how surges in geopolitical uncertainty fatten the tails of the distribution, allowing for a nuanced analysis of vulnerabilities that linear models frequently miss.

To quantify these tail risks and vulnerabilities conditional on changes in geopolitical and macroeconomic regressors, the empirical procedure is structured in sequential steps. Rather than collapsing the complexity of the oil market into a single conditional mean forecast, the analysis begins by estimating multiple quantile regressions at selected points of the conditional distribution, from the lower tail through the median to the upper tail, so that the model can reveal how geopolitical risk influences both the median and the extremes. The raw output from the quantile regressions provides a discrete inverse cumulative distribution function, which is then smoothed by fitting a flexible continuous distribution. Once the continuous conditional distribution is built, the final step evaluates the magnitude of the upside and downside risks, describing how vulnerable the baseline forecast might be to unexpected shocks. This vulnerability is measured using Value-at-Risk, Expected Shortfall, Kullback–Leibler divergence, and distributional skewness, each capturing a distinct dimension of the risk profile.

The remainder of this thesis is organised as follows. Section 2 reviews the relevant literature. Section 3 sets out the contributions. Section 4 describes the data. Section 5 develops the methodological framework, covering quantile regression, the direct forecasting design, the CAViaR augmentation, distributional fitting via the Johnson S_U, and the suite of risk and vulnerability metrics. Section 6 presents the empirical implementation and out-of-sample results. Section 7 studies the entropy-based early warning indicator. Section 8 concludes. The appendices collect the full technical development of every diagnostic and evaluation test, and a Graphical Annex displays the figures referenced throughout.

2 Background and Literature Review

The methodological foundation of this research builds upon the Vulnerable Growth paradigm introduced by Adrian et al. (2019). By modelling the conditional distribution of gross domestic product growth as a function of economic and financial conditions, their framework observed that deteriorating financial conditions tend to increase conditional volatility and shift the lower quantiles downward, while upper quantiles remain comparatively stable. This paradigm indicated that downside risks exhibit significant variance based on financial tightness, highlighting the limitations of conditional mean estimation and the value of evaluating tail vulnerabilities. Extending this rationale suggests that similar non-linear, tail-heavy dynamics might govern other critical macroeconomic variables that are frequently subjected to sudden, severe shocks.

The recent literature has adapted this tail-risk perspective to the effects of geopolitical uncertainty on energy markets. Brignone et al. (2025) examine the non-linear transmission mechanisms of geopolitical risk shocks, suggesting that large positive shocks can trigger a disproportionate amplification channel driven by rising uncertainty. By decomposing geopolitical risk, they observe that geopolitical acts often function similarly to negative demand shocks, whereas geopolitical threats tend to behave as positive uncertainty shocks that elevate oil prices and inflation expectations. Similarly, Verduzco-Bustos and Zanetti (2026) use high-frequency data to isolate geopolitical oil price shocks, finding that these events typically resemble severe supply disruptions, inducing immediate production declines and sharp price increases. They further note that these shocks are coupled with heightened uncertainty that prompts a distinct precautionary inventory response, characterised by an initial surge in inventories followed by a gradual drawdown as the immediate geopolitical threat recedes. Their work reinforces the view that geopolitical risk does not merely shift the conditional mean; it reshapes the distribution of future oil price outcomes.

The quantile regression methodology was proposed by Koenker and Bassett (1978), who introduced the estimation of conditional quantile functions as an alternative to the conditional mean. Their approach provides a distribution-agnostic estimator that is robust to outliers and capable of revealing heterogeneous covariate effects across the conditional distribution. In the context of risk management, quantile regression acquired a dynamic dimension through the Conditional Autoregressive Value-at-Risk (CAViaR) framework of Engle and Manganelli (2004), which allows the conditional quantile to depend autoregressively on its own past values and on past violations, thereby capturing the persistence and clustering of tail events that static models miss.

The distributional fitting step of the *Growth-at-Risk* framework has traditionally relied on the Azzalini skewed-t distribution, as in Adrian et al. (2019). However, the choice of parametric family is not innocuous. The Johnson S_U distribution, introduced by Johnson (1949), offers an alternative with independent control over skewness and kurtosis through two shape parameters, and a closed-form quantile function that eliminates the need for numerical root-finding. These properties make it suited to modelling commodity returns, where the direction and magnitude of tail asymmetry shift as the geopolitical regime evolves. The S_U family has been used in portfolio management and Value-at-Risk contexts, and its application to the distributional step of a *Growth-at-Risk*-type framework is, to our knowledge, uncommon in the energy-finance literature.

The backtesting of conditional risk measures draws on a well-established body of work. Kupiec (1995) proposed the unconditional coverage test that compares the empirical frequency of Value-at-Risk exceedances against the nominal rate. Christoffersen (1998) extended this to a conditional coverage framework that also tests for serial independence of exceedances. The dynamic quantile test of Engle and Manganelli (2004) provides a stringent evaluation by testing whether the hit sequence is a martingale difference sequence. For Expected Shortfall, the backtesting framework of Acerbi and Székely (2014, 2017) exploits the joint elicitability of VaR and ES, constructing tests based on the severity of exceedances rather than merely their frequency. The scoring-rule theory of Gneiting and Raftery (2007) provides the grounding for strictly proper scoring rules used in model comparison. Together, these tools form the backtesting apparatus applied in this thesis. The information-theoretic component of the framework rests on Shannon (1948) and the relative-entropy measure of Kullback and Leibler (1951).

3 Contributions

This thesis makes the following contributions to the literature on commodity market risk and econometric methodology.

First, it introduces the concept of *Oil-at-Risk* (OaR), adapting the *Growth-at-Risk* (GaR) framework of Adrian et al. (2019) to the crude oil market. The adaptation enables probabilistic tail-risk assessments of oil prices in the same language used for macrofinancial surveillance.

Second, it integrates Conditional Autoregressive Value-at-Risk (CAViaR) ideas (Engle and Manganelli, 2004) into the OaR framework. This enrichment supplements the static quantile regression step of the canonical GaR procedure with variables that encode the model's own recent boundary violations, aiming to capture the persistence and asymmetric adjustment of tail risk observed in oil price behaviour.

Third, it departs from the conventional use of the skewed-t distribution for distributional fitting by adopting the Johnson S_U (JSU) family (Johnson, 1949). The JSU system provides a four-parameter characterisation capable of matching a range of skewness and kurtosis combinations, and it is compared here against the skewed-t benchmark within the same pipeline.

Fourth, it examines an entropy-based early warning indicator that exploits daily data. The indicator is the Kullback–Leibler divergence (Kullback and Leibler, 1951) of the fitted conditional density against a reference density, decomposed by tail. Its anticipatory content is assessed with the Cross-Correlation Function and the Granger causality test (Granger, 1969), rather than being assumed.

Fifth, it applies a common backtesting battery uniformly across all model variants. Kupiec unconditional coverage tests (Kupiec, 1995), Christoffersen conditional coverage tests (Christoffersen, 1998), and dynamic quantile tests (Engle and Manganelli, 2004) are applied to every specification, together with Diebold–Mariano comparisons (Diebold and Mariano, 1995).

Sixth, it delivers a self-contained, open-source Python module that implements the full OaR pipeline, covering data preprocessing, quantile regression under both standard and CAViaR-augmented specifications, distributional fitting, entropy computation, and backtesting diagnostics. The module is intended to be publicly available in the author's repository to support replication and extension.

4 Data Description

The primary dependent variable is the European Brent crude oil spot price, measured in U.S. dollars per barrel and sourced from the U.S. Energy Information Administration (via FRED). The daily sample begins on 2 January 1990 and ends on 30 April 2026, providing a historical window that captures numerous major geopolitical crises. Brent is often preferred as a global benchmark because it accounts for a large share of the globally traded crude oil supply and tends to reflect international seaborne market conditions. Its evolution over the sample is shown in Figure **??**. To address non-stationarity, raw prices are transformed into daily log-returns, and weekends and recognised market holidays are removed to maintain a continuous trading-day alignment. The sample median daily log-return is close to zero, a regularity consistent with the near-martingale behaviour of high-frequency commodity returns; this observation matters below, because it implies that the median quantile carries little directional information and that most of the conditioning signal is expected to appear in the tails.

The main explanatory variable is the daily Geopolitical Risk Index (GPRD), an automated, dictionary-based measure that calculates the share of newspaper articles discussing adverse geopolitical events and tensions. The index is entered as a seven-day moving average, a choice motivated in Section 5.2 by the fact that the underlying index is published and interpreted on a calendar-week basis. Country-specific versions may be used to evaluate localised risk origins.

To help ensure that the quantile regression isolates the specific effect of geopolitical risk, a targeted set of exogenous control variables is incorporated. The U.S. Dollar Index tracks the relative strength of the United States dollar against a basket of major foreign currencies. Because Brent crude is priced in dollars, an appreciation of the currency tends to suppress global demand and lower the commodity's nominal price. Including this variable helps prevent exchange-rate dynamics from being conflated with geopolitical shocks. The series is expressed as a log-difference to enforce stationarity and to capture immediate exchange-rate elasticity.

Realised volatility measures the magnitude of recent price dislocations, controlling explicitly for volatility clusters and latent heteroskedasticity. Constructed from the historical variation of crude returns, this metric serves as a barometer for the baseline level of underlying market risk. Controlling for realised volatility helps purge the residual variance of financial panic, isolating the structural pricing of the geopolitical threat itself. The handling of this variable differs between the in-sample specification and the out-of-sample forecasting exercise, a distinction discussed in Sections 5.2 and 5.4.

The Baltic Dry Index (BDI) is a daily composite index measuring the average cost of transporting major raw materials by sea. The BDI serves as a real-time proxy for global industrial demand and real economic activity. Because it tracks physical shipping constraints rather than financial derivatives, it is largely free of speculative noise. Including the BDI controls for fundamental demand-side pressures, helping the model separate geopolitical shocks from standard macroeconomic business cycles. The daily values are transformed into log-differences to capture the high-frequency growth rate of global shipping demand and to enforce stationarity.

5 Methodology

When examining the effect of geopolitical risk on the rate of change of Brent crude prices, traditional linear methodologies reveal their limitations. An initial look at the data displays a high presence of influential observations, data points located far outside the dense central cloud that can heavily distort OLS regressions. Because OLS minimises the sum of squared residuals, it is highly sensitive to these extreme outliers, and the resulting slope coefficients often collapse close to zero, offering little predictive power. Linear methods focus entirely on the expected value or conditional mean, ignoring parts of the distribution such as the tails. The *Growth-at-Risk* methodology, adapted here as *Oil-at-Risk*, provides a direct econometric alternative. By isolating specific quantiles, it describes how surges in geopolitical uncertainty fatten the tails of the distribution, enabling a nuanced analysis of vulnerabilities that linear models miss.

5.1 Quantile Regression

While standard OLS regression provides a powerful tool for estimating the Conditional Expectation Function, it operates under strict assumptions and is highly sensitive to outliers. Quantile regression, introduced by Koenker and Bassett (1978), estimates the Conditional Quantile Function (CQF). Instead of modelling the mean, it models a specific quantile τ (for instance, the 10th or 90th percentiles), making it distribution-agnostic and robust to extreme outliers.

The CQF, given a vector of regressors X_i, is the inverse of the conditional cumulative distribution:

$$Q_\tau(Y_i \mid X_i) = F_Y^{-1}(\tau \mid X_i). \tag{1}$$

Just as OLS derives the conditional mean by minimising the expected squared error, quantile regression derives the CQF by minimising an expected asymmetric loss:

$$Q_\tau(Y_i \mid X_i) = \underset{q(X_i)}{\arg\min}\ \mathbb{E}[\rho_\tau(Y_i - q(X_i))]\,, \tag{2}$$

where $\rho_\tau(u)$ is the check function. To trace out specific percentiles of a distribution, the regression must penalise under-predictions and over-predictions asymmetrically. If we are estimating the 90th quantile ($\tau = 0.90$), the model should heavily penalise under-predictions (negative residuals) and lightly penalise over-predictions (positive residuals). To achieve this while maintaining the convexity required for optimisation, a piecewise linear function based on the absolute error is constructed. Multiplying the positive piece by the target quantile τ and the negative piece by $(1 - \tau)$ introduces the necessary asymmetry.

The check function is written as:

$$\rho_\tau(u) = u \cdot \Big(\tau - \mathbf{1}_{u<0}\Big), \tag{3}$$

where $\mathbf{1}_{u<0}$ is an indicator function taking the value 1 if the residual is negative and 0 otherwise. Defining the residual as $u_t = y_t - x_t'\beta_\tau$, the objective is to find the parameter vector $\hat{\beta}_\tau$ that minimises the sum of this asymmetric check function across all observed periods:

$$\hat{\beta}_\tau = \underset{\beta_\tau \in \mathbb{R}^k}{\arg\min} \sum_{t=1}^{T} \rho_\tau(y_t - x_t'\beta_\tau)\,. \tag{4}$$

The economic appeal of this estimator is immediate. In the context of crude oil markets, OLS would estimate a single slope linking geopolitical risk to average returns. By contrast, quantile regression estimates a separate slope at each quantile level, revealing whether geopolitical shocks predominantly compress the lower tail (raising downside vulnerability), stretch the upper tail (amplifying upside spikes), or shift the entire distribution. This heterogeneity across quantiles is the identifying premise of the *Oil-at-Risk* framework. The optimisation procedure, which transforms the non-differentiable check function into a standard linear programme, is detailed in Appendix A.

5.2 Empirical Specification

The empirical specification retained for in-sample estimation regresses daily Brent crude oil log-returns on a seven-day moving average of the Geopolitical Risk Index and a vector of control variables. For each quantile in the target set $\tau \in \{0.01, 0.05, 0.25, 0.50, 0.75, 0.95, 0.99\}$, the empirical equation is:

$$Q_\tau(\Delta\text{Brent}_t) = \beta_0(\tau) + \beta_1(\tau)\,\text{GPRD-MA7}_t + \gamma(\tau)'X_t, \tag{5}$$

where the in-sample control vector is:

$$X_t = \Big(\Delta\text{USDI}_t,\ \text{RV}_{t-1},\ \Delta\text{BDI}_t\Big)'. \tag{6}$$

A single specification is imposed uniformly across all quantiles to preserve comparability across the conditional distribution and to prevent tail-specific overfitting. The estimated coefficient paths across τ, together with the OLS benchmark, are displayed in Figure **??**.

The primary regressor is a seven-day moving average of the Geopolitical Risk Index, chosen to match a calendar week of the underlying index, which is compiled on a calendar-week basis (seven calendar days):

$$\text{GPRD-MA7}_t = \frac{1}{7}\sum_{j=0}^{6} \text{GPRD}_{t-j}. \tag{7}$$

The window is calibrated to the risk-transmission mechanism operating through financial markets. Geopolitical shocks rarely resolve within a single trading session; their propagation from the news cycle to market positioning typically unfolds over several days as risk premia adjust, hedging demand builds, and institutional investors reassess exposure. Smoothing over seven calendar days therefore captures the sustained escalation of international tensions while filtering the transient spikes generated by individual headlines, so that $\hat{\beta}_1(\tau)$ reflects structural geopolitical stress rather than short-lived media events.

Realised volatility enters the in-sample regression at lag $t-1$ rather than contemporaneously, as a deliberate precaution against look-ahead bias. Because realised volatility at time t is computed from intraperiod price movements, it is partly determined by the dependent variable itself, which would induce a simultaneity problem in a contemporaneous ($h = 1$) regression. Using RV_{t-1} ensures the regressor is fully observable before period t opens. This concern is specific to the in-sample specification, in which the target and the volatility proxy share the same date. In the out-of-sample direct forecasting exercise of Section 5.4, where the target is the return realised h trading days ahead, contemporaneous realised volatility RV_t is already part of the information set available at the forecast origin and cannot leak future information; the lag is therefore unnecessary there, and RV_t is used.

Equation (5) represents the parsimonious specification selected after screening a wider set of candidate regressors. Among the alternatives evaluated but not retained were measures of real economic activity (industrial production indices and PMI composites), weekly U.S. crude oil supply and production data from the Energy Information Administration, and the Henry Hub natural gas spot price. These variables were excluded on grounds of multicollinearity with the retained controls, insufficient within-sample variation at the daily frequency, or failure to improve out-of-sample quantile score performance.

5.3 Diagnosis of the Quantile Regression

The diagnostic battery provides broad empirical support for the validity of specification (5). The Koenker–Machado $R^1(\tau)$ (Koenker and Machado, 1999) indicates that the inclusion of the regressors yields a meaningful reduction in the weighted objective function relative to the intercept-only benchmark, with explanatory power concentrated in the tail quantiles, precisely where the *Oil-at-Risk* metric is most informative. This concentration is expected rather than anomalous: the conditioning variables are designed to describe extreme events, and, because the median daily return is close to zero and carries little directional signal, the regressors add proportionally more at the tails than at the centre. The QAR unit root test of Koenker and Xiao (2004) rejects the null of quantile-specific non-stationarity at all targeted quantiles, consistent with mean-reverting behaviour of Brent log-returns throughout the conditional distribution. The ARCH-LM test (Engle, 1982) fails to reject the null of no residual autoregressive conditional heteroskedasticity at conventional significance levels, indicating that the specification has largely absorbed the volatility clustering inherent in daily crude oil returns. The Wald test for equality of slopes across quantiles rejects the null of coefficient constancy, supporting the view that the regressors exert heterogeneous effects across the conditional distribution, which is the identifying premise of the *Oil-at-Risk* framework. No quantile crossing violations are detected in the fitted paths, consistent with the monotonicity requirement across all observations in the sample. Full tabulations of all test statistics, degrees of freedom, and associated p-values are reported in Appendix B.

Among these diagnostics, the Dynamic Quantile test of Engle and Manganelli (2004) warrants particular attention. The test defines the centred hit sequence:

$$\mathrm{Hit}_t(\tau) = \begin{cases} 1-\tau & \text{if } y_t < \hat{q}_t(\tau), \\ -\tau & \text{if } y_t \geq \hat{q}_t(\tau), \end{cases} \tag{8}$$

Under correct specification, violations must be serially uncorrelated and independent of all information available at $t-1$. The null hypothesis is $H_0\colon \mathbb{E}[\mathrm{Hit}_t(\tau) \mid \Omega_{t-1}] = 0$. The hits are regressed via OLS on an instrument matrix Z_t comprising a constant, lagged values of $\mathrm{Hit}_t(\tau)$, and lagged fitted quantiles. The test statistic,

$$DQ = N \cdot R^2 \xrightarrow{d} \chi^2_q, \tag{9}$$

is asymptotically distributed as χ^2 with q degrees of freedom equal to the number of instruments. Rejection signals residual predictability in the hit sequence, indicating that the specification fails to absorb relevant tail dynamics.

5.4 Direct Multi-Step Forecasting

The goal of the out-of-sample analysis is genuinely predictive: to anticipate the conditional distribution of Brent returns several periods into the future, using only the information available today. There are two classical ways to produce such multi-step forecasts. The iterated (recursive) approach fits a single one-step model and chains it forward, so that a two-step forecast is obtained by feeding the one-step forecast back into the same equation. This is fragile in practice: any misspecification in the one-step model compounds as the horizon grows, and, more decisively for the present setting, a quantile cannot be iterated at all, because the quantile of an expectation is not the expectation of a quantile.

The analysis therefore adopts direct forecasting. Instead of chaining a one-step model forward, a separate quantile regression is estimated for each horizon h by shifting the target back by h periods, so that the information available at t is paired with the realised future value Y_{t+h}:

$$\hat{\beta}^{(h)}(\tau) = \arg\min_{\beta} \sum_{t=1}^{T-h} \rho_\tau(Y_{t+h} - X_t'\beta)\,, \quad Q_\tau(Y_{t+h} \mid X_t) = X_t'\beta^{(h)}(\tau). \tag{10}$$

Because a dedicated model is fitted at each horizon, the estimator never iterates forward and is immune to the compounding bias of the recursive scheme; it targets the horizon directly. This design also lets us study how the geopolitical risk premium propagates over time by re-estimating equation (10) over a grid of horizons h and observing how the conditional quantiles, and their predictive accuracy, evolve as h increases. It is in this setting that contemporaneous realised volatility RV_t may be used without look-ahead bias, as noted in Section 5.2, because the target lies h trading days beyond the forecast origin.

To evaluate out-of-sample predictive ability, the information used to estimate the model must be kept strictly apart from the information used to score it. The model is estimated on a fixed rolling window of 1,008 trading days (four trading years, in line with the Basel III internal-model calibration; Basel Committee on Banking Supervision, 2019). At each forecast origin t, the model trains on $[t-1008,\, t)$, issues a single h-step forecast, and the window advances one day. Sweeping across every eligible origin, this sequence yields a statistically meaningful measure of predictive capability rather than the artefact of a single origin. The fixed window also lets the parameters continuously forget superseded regimes and re-anchor on the prevailing one, which matters in a market whose volatility structure is reshaped by every new geopolitical episode. The scatter of realised returns against the forecasted quantile envelope is shown in Figure **??**, and the behaviour of the in-sample forecast around a stressed episode is illustrated in Figure **??**.

5.5 Competing Specifications: Capturing Geopolitical Tail Risk

The standard quantile regression is an accurate description of the typical conditional quantile, but it is, in a precise sense, memoryless about its own failures. It maps today's geopolitical and financial covariates into a quantile forecast and then forgets whether that forecast was breached. In tranquil markets this is harmless; in the tails it is not.

The economic object of interest is not the centre of the distribution but its extreme edge, the threshold beyond which an oil position suffers a large loss. Extreme oil moves are not independent draws: they cluster. A supply shock threatening the Strait of Hormuz, or a demand collapse of the COVID-19 type, does not resolve in a single session; it propagates through inventory decisions, futures-curve repositioning, and risk re-pricing over days and weeks. Empirically this shows up as volatility clustering in the tail, where a Value-at-Risk breach today materially raises the probability of another breach tomorrow. A model that treats each tail event as an isolated surprise is therefore caught off guard precisely when accuracy matters most.

This is the gap that Engle and Manganelli (2004) address with the CAViaR framework, in which the conditional quantile is allowed to depend autoregressively on its own past behaviour, so that the tail remembers recent stress. The present thesis does not estimate a full CAViaR recursion; instead it imports the central insight into the direct-forecasting design by augmenting the standard specification with variables that encode the model's own recent boundary violations. Two augmentation variants are considered.

5.5.1 Model 2: CAViaR Indicator Augmentation

The first variant appends a pair of binary stress flags to the regressor set:

$$\text{upside\_breach}_t = \mathbf{1}\Big(y_t > Q_{\text{high}}(y_t \mid x_{t-h})\Big), \tag{11}$$

$$\text{downside\_breach}_t = \mathbf{1}\Big(y_t < Q_{\text{low}}(y_t \mid x_{t-h})\Big). \tag{12}$$

Each flag fires if the most recent realised return exceeded the model's own upper or lower boundary, defined here at the $\{0.05, 0.95\}$ breach quantiles. A positive coefficient on these indicators means that the model widens its predicted quantile corridor after a breach, absorbing the persistence that a memoryless specification would miss.

5.5.2 Model 3: CAViaR Severity Augmentation

A binary flag records that a breach happened but is blind to its violence: a 1% breach and a 10% breach speak with the same voice. Yet for tail risk the magnitude is much of the story, the difference between a manageable loss and a blow-up. The severity variant therefore replaces the indicators with the absolute size of the exceedance, the fallout error:

$$\text{upside\_severity}_t = \max\Big(0,\ y_t - Q_{\text{high}}(y_t \mid x_{t-h})\Big) \geq 0, \tag{13}$$

$$\text{downside\_severity}_t = \max\Big(0,\ Q_{\text{low}}(y_t \mid x_{t-h}) - y_t\Big) \geq 0. \tag{14}$$

Entering the fallout error in absolute value forces its coefficient to act as a pure stress-intensity loading, uncontaminated by the sign of the exceedance, so the upper- and lower-tail severities read as independent measures of how hard the market overshot. Because both augmentations need a realised value to define a breach, the stress signal they carry is sharp at very short horizons but decays as h lengthens, an economic limitation the out-of-sample results make concrete.

5.6 Distribution Fitting: From Discrete Quantile Forecasts to a Continuous Conditional Density

The direct forecasting model delivers, for every forecast origin t and horizon h, a finite set of predicted conditional quantiles:

$$\hat{Q}_{t+h} = \left\{\hat{q}_{t+h}(\tau_1),\ \ldots,\ \hat{q}_{t+h}(\tau_K)\right\}, \quad \tau_k \in \{0.01,\, 0.05,\, 0.25,\, 0.50,\, 0.75,\, 0.95,\, 0.99\}. \tag{15}$$

These point forecasts are the discrete skeleton of the conditional distribution of future Brent returns. A portfolio manager who needs only a single Value-at-Risk threshold at the 5th percentile can read $\hat{q}_{t+h}(0.05)$ directly and stop. But we would like to extend the *Oil-at-Risk* framework well beyond a single threshold.

Consider what a central bank or a sovereign wealth fund actually needs when a geopolitical crisis unfolds in the Persian Gulf. The question is not merely whether the 5th percentile of the return distribution has moved; it is how the entire shape of the conditional density has changed. Has the left tail fattened while the right tail remained stable, signalling a pure downside vulnerability? Has skewness reversed, as it does when a demand-side collapse overtakes a supply-side fear? To compute Expected Shortfall, one must integrate below the quantile. To measure Kullback–Leibler divergence between successive conditional densities, one must have a density. To study how geopolitical shocks reshape the skewness and kurtosis of the forecast distribution across horizons, one must recover the full shape of the curve, not merely a handful of points on it. The discrete quantile skeleton must therefore be interpolated into a continuous parametric density that preserves the information encoded in the quantile forecasts while supplying the analytic tractability that downstream tail-risk metrics require.

5.6.1 The Johnson S_U Distribution

The choice of parametric family must satisfy two requirements. First, it must be flexible enough to accommodate the stylised facts of crude oil returns: excess kurtosis, asymmetric tails, and time-varying skewness whose direction shifts as the geopolitical regime evolves. When a supply disruption threatens, the right tail of the return distribution fattens as the market prices in the possibility of a sharp price spike; when a demand collapse materialises, the left tail fattens instead. A distribution that imposes symmetry, or that controls tail weight through a single parameter, cannot track these asymmetric rotations. Second, its cumulative distribution function must admit a closed-form inverse, because the estimation procedure evaluates the theoretical quantile function many thousands of times.

The Johnson S_U distribution satisfies both. A random variable Y follows a Johnson S_U if:

$$Y = \xi + \lambda \sinh\left(\frac{Z - \gamma}{\delta}\right), \quad Z \sim \mathcal{N}(0, 1), \tag{16}$$

where ξ is the location parameter, $\lambda > 0$ is the scale, γ governs skewness, and $\delta > 0$ governs tail weight. The quantile function follows by replacing Z with $\Phi^{-1}(p)$:

$$Q(p) = \xi + \lambda \sinh\left(\frac{\Phi^{-1}(p) - \gamma}{\delta}\right). \tag{17}$$

This closed-form quantile function is the decisive computational advantage: it evaluates rapidly, involves no numerical root-finding, and makes the Minimum Distance Estimator described below as fast as a simple least-squares problem.

The economic appeal of the S_U family becomes clear when one considers the alternative. The Generalised Error Distribution, widely used in GARCH modelling (Nelson, 1991), controls tail weight through a single shape parameter but imposes symmetry: it cannot distinguish between a left-skewed distribution during a demand collapse and a right-skewed one during a supply panic. The Johnson S_U provides independent control over skewness and kurtosis through the two separate shape parameters γ and δ. In the limit $\gamma \to 0$, $\delta \to \infty$, the S_U nests the Gaussian as a special case, so that tranquil periods are accommodated without forcing heavy tails where none exist. The original *Growth-at-Risk* framework of Adrian et al. (2019) employs the Azzalini skewed-t distribution for this step; the S_U is a more finance-oriented alternative that has been used to model asset returns in portfolio management and Value-at-Risk contexts, with comparable flexibility and a simpler quantile-function inversion.

5.6.2 Minimum Distance Estimation

In the classical Maximum Likelihood setting, the analyst observes many realisations from the return-generating process, each an independent draw from the stochastic process whose density is to be estimated. The likelihood surface is well identified, the sample is large, and the estimator is efficient. The setting that arises after the quantile regression step is different. For a given forecast origin t and horizon h, we do not observe a large sample of future returns. We observe a small vector of K predicted quantiles, one for each targeted quantile level. With $K = 7$, we have seven data points, not seven thousand. The natural estimator in this setting is the Minimum Distance Estimator (MDE), which finds the parameter vector $\Theta = (\xi, \lambda, \gamma, \delta)$ that makes the theoretical quantile function of the chosen distribution pass as close as possible to the predicted quantiles:

$$\hat{\Theta}_t = \arg\min_{\Theta} \sum_{k=1}^{K} \Big[\hat{q}_{t+h}(\tau_k) - Q_{\mathrm{JSU}}(\tau_k;\, \Theta)\Big]^2. \tag{18}$$

Two algorithmic choices materially affect the reliability of the solution. First, the initial guess Θ_0 strongly influences the speed and likelihood of convergence. The initialisation sets ξ at the predicted median $\hat{q}_{t+h}(0.50)$, λ at the interquartile range $\hat{q}_{t+h}(0.75) - \hat{q}_{t+h}(0.25)$, and γ and δ at zero and one, respectively. This encodes a symmetric, moderately heavy-tailed prior centred on the regression's median forecast, a sensible starting point that the optimiser then refines as it absorbs the asymmetry embedded in the outer quantiles. Second, convergence tolerances must be tight enough to produce an accurate fit but loose enough to avoid excessive iterations on nearly flat regions of the objective (sum-of-squares) surface; a function tolerance of 10^{-10} has proven adequate across the full out-of-sample experiment.

The procedure is repeated independently for every forecast origin in the rolling window, yielding a time series of fitted parameter vectors $\{\hat{\Theta}_t\}_{t\in W}$ and, through them, a time series of continuous conditional densities. Each density is anchored to the information set available at its forecast origin: it inherits the conditioning embedded in the quantile forecasts and imposes no cross-sectional pooling across dates. The result is a sequence of distributions that evolves with the geopolitical and macroeconomic environment, fattening its left tail when the Geopolitical Risk Index spikes and narrowing it when conditions normalise, which is the object the *Oil-at-Risk* framework requires. An in-sample example of a fitted conditional density is displayed as a three-dimensional surface in Figure **??**.

5.6.3 Evaluation: The Probability Integral Transform

Once the MDE has produced a fitted conditional distribution for every forecast origin, the critical question becomes whether those distributions are accurate descriptions of the future reality they purport to model. The foundational diagnostic is the Probability Integral Transform (PIT), formalised in the context of density forecast evaluation by Diebold et al. (1998). For each forecast origin t in the out-of-sample window, the PIT evaluates the realised return y_{t+h} under the conditional CDF that was forecasted for that date:

$$u_t = F\Big(y_{t+h} \mid \hat{\Theta}_t\Big)\,. \tag{19}$$

If the fitted distribution is the true conditional distribution, the transformed value u_t is exactly Uniform$(0, 1)$:

$$P(U \leq u) = u \quad \text{for all } u \in [0, 1]. \tag{20}$$

The contrapositive supplies the diagnostic: if the sequence of PITs departs from uniformity, the forecasted distribution is misspecified. Departures manifest in characteristic ways that carry direct economic meaning. An excess of PITs near zero indicates that the model's left tail is too thin, understating the probability of large negative returns, precisely the kind of miscalibration that would lead a risk manager to understate her Expected Shortfall during a geopolitical crisis. The PIT histogram and the corresponding diagonal (empirical-versus-theoretical CDF) plot are shown in Figures **??** and **??**.

The formal assessment is provided by the Kolmogorov–Smirnov (KS) test (Massey, 1951), which measures the supremum absolute deviation between the empirical CDF of the PIT sequence and the theoretical CDF of the Uniform$(0, 1)$ distribution:

$$D_n = \sup_{u \in [0,1]} \left| \hat{F}_n(u) - u \right|. \tag{21}$$

The PIT evaluation is conducted over the same out-of-sample window used for the coverage backtests. This design constitutes a single, continuous goodness-of-fit assessment that spans the entire out-of-sample period, evaluating every fitted density across every forecast origin simultaneously. It is more demanding than a pointwise coverage backtest, because it asks whether the entire shape of the conditional density, its location, its spread, its skewness, and its tail weight, is correctly specified at every date. A PIT sequence consistent with uniformity is therefore a necessary condition for treating the fitted distributions as a credible probabilistic characterisation of future oil price vulnerabilities under geopolitical duress.

5.7 Risk and Vulnerability Metrics

Having recovered a continuous conditional Johnson S_U density at every forecast origin, the framework extracts the risk and vulnerability metrics that constitute the diagnostic output of the *Oil-at-Risk* framework. A density, by itself, is not a decision tool. A risk manager facing a potential supply disruption in the Strait of Hormuz does not consult a probability density function; she asks how large her worst-case loss might be, how severe the tail would be if that worst case materialises, and whether the current distribution has shifted in a way that signals an abnormal accumulation of downside risk relative to historical norms. These are different questions, and they require different metrics.

5.7.1 Value-at-Risk

Value-at-Risk is the α-quantile of the conditional return distribution. For the left tail:

$$\mathrm{VaR}_{t+h}^{\mathrm{Left}}(\alpha) = Q_\alpha\Big(r_{t+h} \mid \hat{\Theta}_t\Big), \tag{22}$$

and for the right tail:

$$\mathrm{VaR}_{t+h}^{\mathrm{Right}}(1-\alpha) = Q_{1-\alpha}\Big(r_{t+h} \mid \hat{\Theta}_t\Big), \tag{23}$$

where $\mathrm{VaR}^{\mathrm{Left}} < 0$ represents a loss threshold and $\mathrm{VaR}^{\mathrm{Right}} > 0$ represents a gain threshold. In plain terms, the left-tail measure reports the return the market would realise under a very adverse scenario, that is, a return so low that it is expected to be exceeded on the downside only with probability α; the right-tail measure reports the return under a very favourable scenario, so high that it is expected to be exceeded on the upside only with probability α. Computing both is essential because the conditional distribution fitted by the Johnson S_U is generically asymmetric: when the Geopolitical Risk Index is elevated and the dollar is weakening, the two tails need not move in parallel, and reporting only the left tail would discard the information contained in the right tail's response to the same conditioning variables.

The conditional VaR follows directly from the fitted density:

$$\mathrm{VaR}_{t+h}^{\mathrm{Cond,\ Left}}(\alpha) = \hat{\xi}_t + \hat{\lambda}_t \sinh\left(\frac{\Phi^{-1}(\alpha) - \hat{\gamma}_t}{\hat{\delta}_t}\right). \tag{24}$$

A useful observation is that no time-scaling is needed for the conditional VaR. The quantile regression that produced the inputs to the distribution fitting step was estimated with the target variable y_{t+h}, the return over the next h trading days. The fitted Johnson S_U distribution is therefore already a distribution of h-step-ahead returns. Its α-quantile is directly the h-day VaR, without any square-root-of-time adjustment.

The unconditional VaR, computed by historical simulation over a rolling window of $W = 1{,}008$ observations (four trading years, following Basel III guidelines; Basel Committee on Banking Supervision, 2019), serves as the benchmark against which the conditional measure is assessed. When the two VaR series track each other closely, the conditioning variables are adding little information beyond what the recent history already contains. When they diverge, the conditional model is detecting a regime shift that the backward-looking historical measure has not yet absorbed. The conditional and unconditional VaR and CVaR series are displayed in Figure **??**.

5.7.2 Expected Shortfall

Value-at-Risk answers how bad a loss can be at a given confidence level, but it is silent on what happens beyond that threshold. Two distributions can share the same 97.5th percentile while having radically different tail behaviour beyond it: one may decay rapidly to zero, implying that exceedances beyond the VaR are rare and mild, while the other may exhibit a heavy tail extending far into the loss region, implying that when the VaR is breached, the resulting loss is likely to be large. In crude oil markets, this distinction is not hypothetical. The distribution of returns during the COVID-19 demand collapse of April 2020 had a left tail extending to losses of 30 percent or more in a single day, while the distribution during the preceding months of gradual price decline had a much thinner left tail despite a similar VaR at the 2.5th percentile.

Expected Shortfall addresses this limitation by measuring the average loss in the tail beyond the VaR threshold:

$$\mathrm{ES}_{t+h}^{\mathrm{Left}}(\alpha) = \mathbb{E}\Big[r_{t+h} \mid r_{t+h} \leq \mathrm{VaR}_{t+h}^{\mathrm{Left}}(\alpha)\Big]\,. \tag{25}$$

The left-tail Expected Shortfall is always more negative than the left-tail VaR, because the conditional expectation in the tail lies beyond the tail's boundary by construction. The difference $\mathrm{ES}^{\mathrm{Left}} - \mathrm{VaR}^{\mathrm{Left}}$ measures the expected severity of a VaR exceedance, a quantity of direct interest to a risk manager who has already accepted that the VaR threshold may be breached and wants to know how bad the outcome is likely to be.

Expected Shortfall enjoys a property that Value-at-Risk does not: it is a coherent risk measure in the sense of Artzner et al. (1999). The critical property is sub-additivity, which was one of the principal motivations behind the Basel Committee's decision, in the Fundamental Review of the Trading Book, to replace VaR with Expected Shortfall as the basis for internal-model capital calculations under Basel III (Basel Committee on Banking Supervision, 2019).

When the conditional density is available in closed form, the Expected Shortfall integral is evaluated by numerical integration on a fine grid of 5,000 points spanning the support of the distribution:

$$\mathrm{ES}_{t+h}^{\mathrm{Cond,\ Left}}(\alpha) \approx \frac{\displaystyle\sum_{x_i \leq \mathrm{VaR}^{\mathrm{Left}}} x_i\, f(x_i \mid \hat{\Theta}_t)\,\Delta x}{\displaystyle\sum_{x_i \leq \mathrm{VaR}^{\mathrm{Left}}} f(x_i \mid \hat{\Theta}_t)\,\Delta x}. \tag{26}$$

5.7.3 Kullback–Leibler Divergence as a Vulnerability Indicator

Value-at-Risk and Expected Shortfall are point statistics extracted from the conditional density at specific tail probabilities. They answer questions about the boundary and depth of the tails, but they do not directly measure how much the conditional density has departed from a reference state. Kullback–Leibler divergence (Kullback and Leibler, 1951) provides a global measure of distributional distance that captures precisely this distinction. Given a conditional forecast density $p(x)$ and a reference density $q(x)$:

$$D_{\mathrm{KL}}(p \,\|\, q) = \int_{-\infty}^{\infty} p(x) \ln\left(\frac{p(x)}{q(x)}\right) dx. \tag{27}$$

The divergence is always non-negative and equals zero if and only if the two densities are identical. Its asymmetry is economically meaningful: $D_{\mathrm{KL}}(p \,\|\, q)$ heavily penalises the reference density q when it assigns near-zero probability to events that the conditional density p considers plausible. In risk management terms, this is the dangerous scenario: the reference model says a large loss is essentially impossible, while the conditional forecast says it has non-negligible probability.

The KL divergence can be interpreted through the lens of information theory (Shannon, 1948) as the expected excess surprise when the reference density q is used to describe a world that is actually governed by p:

$$D_{\mathrm{KL}}(p \,\|\, q) = \mathbb{E}_p[-\ln q(x)] - \mathbb{E}_p[-\ln p(x)]\,. \tag{28}$$

The first term is the cross-entropy of q under p; the second is the entropy of p itself. The divergence measures the extra information, in nats, needed to describe the true state of the world when the wrong model is assumed.

Following Adrian et al. (2019), the divergence is decomposed into left-tail, centre, and right-tail components by restricting the integration domain. With $\tau = 0.95$ as the default tail mark (isolating the extreme 5 percent of each tail, rather than the median split used in the original *Growth-at-Risk* framework, which is better suited to quarterly GDP growth than to daily commodity returns):

$$D_{\mathrm{KL}}^{\mathrm{Left}}(p \,\|\, q) = \int_{-\infty}^{q_{1-\tau}} p(x) \ln\left(\frac{p(x)}{q(x)}\right) dx, \tag{29}$$

$$D_{\mathrm{KL}}^{\mathrm{Right}}(p \,\|\, q) = \int_{q_\tau}^{\infty} p(x) \ln\left(\frac{p(x)}{q(x)}\right) dx. \tag{30}$$

Two reference baselines are employed. The first is a Normal density calibrated to the historical mean and standard deviation, whose divergence measures the departure from Gaussianity and functions as an early warning: a rapid increase signals that the market's return distribution is moving away from the well-behaved regime in which traditional risk metrics are reliable. The second is the unconditional Johnson S_U density fitted by Maximum Likelihood to the historical returns, which is the baseline used in the original Vulnerable Growth framework and is shown in Figure **??**. This divergence is more demanding, because the unconditional S_U already accommodates skewness and heavy tails; a high divergence against it signals a genuine regime shift in the conditional distribution, not merely a departure from Gaussianity.

5.7.4 Distributional Skewness

The Fisher skewness coefficient of the fitted Johnson S_U distribution is computed analytically using the closed-form expressions derived by Johnson (1949):

$$\gamma_1 = -\frac{\sqrt{\omega}\,(\omega+2)\left[3\sinh\!\left(\frac{\gamma}{\delta}\right)+\sinh\!\left(\frac{3\gamma}{\delta}\right)\right]}{4\left[\frac{\omega-1}{2}\left(\omega\cosh\!\left(\frac{2\gamma}{\delta}\right)+1\right)\right]^{3/2}}, \tag{31}$$

where $\omega = \exp(1/\delta^2)$. The skewness is negative when the left tail is heavier than the right, signalling an asymmetric accumulation of downside risk, and positive when the right tail dominates. The economic content of time-varying skewness in crude oil markets is substantial. During periods of geopolitical escalation, the right tail fattens relative to the left and skewness becomes positive or less negative. During demand-side recessions, the opposite pattern emerges. Tracking conditional skewness over time, as in Figure **??**, reveals the direction in which the balance of risks is tilted at each forecast origin.

5.8 Computational Configuration

The risk metrics are computed over an out-of-sample evaluation that traverses every trading day in the test span, producing a time series of conditional and unconditional risk measures at each forecast origin.

The forecast horizon h is set to $h = 5$ trading days, corresponding to one trading week, and is used consistently across the direct forecasting, risk-metric, and early-warning steps. This choice reflects the objective of capturing early dynamics in oil prices: a one-week horizon is short enough to retain meaningful predictive content in the quantile regression, yet long enough to provide an operational anticipation window for risk management. Shorter horizons ($h = 1$ or $h = 2$) sharpen the CAViaR stress signal but leave little room for a response, while longer horizons ($h = 22$ or more) inflate the noise in the quantile regression coefficients, which the MDE step then inherits. The one-week horizon also aligns the forecasting, risk-metric, and entropy computations on a single, interpretable time scale.

The confidence level for Value-at-Risk and Expected Shortfall is set at 97.5 percent, following the Basel III Fundamental Review of the Trading Book (Basel Committee on Banking Supervision, 2019). The earlier Basel I and Basel II.5 frameworks used a 99 percent confidence level, but the Basel III revision lowered it to 97.5 percent. Under heavy-tailed distributions, which crude oil returns notoriously exhibit, estimating the 99th percentile with precision requires very large samples. The *Oil-at-Risk* framework mitigates this by fitting the quantile regression at moderate quantile levels, specifically $\tau \in \{0.01, 0.05, 0.25, 0.50, 0.75, 0.95, 0.99\}$, recovering a continuous density via the Johnson S_U, and then reading the 2.5th percentile from the fitted density. The density interpolates between the estimated quantiles, extracting tail information from the shape of the distribution rather than from a single regression at an extreme quantile level.

The rolling window for the historical VaR and ES is set at $W = 1{,}008$ observations, corresponding to four trading years, matching the direct-forecasting training window and the standard calibration window prescribed by Basel III for internal-model VaR calculations. The quantile regression coefficients that underpin the conditional density are refitted once every 30 out-of-sample observations, approximately one trading month, and held fixed in between. On the intervening days, the cached regression coefficients are applied to the current day's predictor values, so the conditional density still updates daily through the changing covariates; only the mapping from covariates to quantiles is held constant between retraining dates. A robustness check with daily retraining (retrain_after = 1) confirms that the risk metric series are nearly indistinguishable, supporting the cadence choice.

6 Empirical Implementation and Out-of-Sample Results

6.1 Evaluating the Forecasts

Three complementary lenses judge the forecasts: a strictly proper loss measured over the rolling window, a magnitude-of-exceedance audit, and a pair of formal coverage backtests. The mechanics of every test are derived in full in Appendix C; here the treatment is kept brief and focused on what each test certifies. All figures below are computed at the operative horizon $h = 5$ and, where a tail is specified, at $\tau = 0.95$.

6.1.1 The Pinball Loss and the Generalisation Gap

A quantile cannot be scored with the symmetric mean-squared error, because over- and under-prediction must be penalised by different amounts. The asymmetric pinball (check) loss supplies that asymmetry:

$$L_\tau(e) = e \cdot \Big(\tau - \mathbf{1}_{e<0}\Big), \quad e = Y_{t+h} - \hat{y}_{t+h|t}(\tau). \tag{32}$$

It is the strictly proper scoring rule for quantiles (Gneiting and Raftery, 2007): it is minimised if and only if the forecast equals the true conditional quantile, so a lower out-of-sample pinball loss is strong evidence of a better tail estimate. Tracking the loss on the training data against the loss on the test data across a dense grid of horizons yields the generalisation gap, the shaded area between the two loss curves, shown in Figure **??**. A small, stable gap indicates that the model captures structural signal; a gap that widens sharply with the horizon is the classic signature of overfitting. In the present experiment the gap stays bounded for the benchmark across the horizons examined, whereas the CAViaR variants begin to separate at longer horizons, consistent with their stale-information handicap.

6.1.2 Fallout (Severity) Audit

Coverage frequency alone can hide catastrophic single-day failures, so the magnitude of every breach is also audited: how far the realised return overshot the predicted 1%–99% tunnel. For risk management, frequent but tiny breaches are far preferable to rare but massive ones, since the latter betray a structural failure to price an impending shock. At $h = 5$ the three models keep the realised return inside the predicted envelope on 97.83% (Standard), 97.17% (CAViaR-indicator) and 97.26% (CAViaR-severity) of test days, with no isolated catastrophic overshoot. The full breach paths for the three models, including the behaviour of the tunnel during the 2020 crisis, are collected in Figure **??**.

6.1.3 Coverage Backtests

Because the model's output is an estimated quantile that functions as a Value-at-Risk threshold, its adequacy is judged with risk-management backtests built on the binary hit sequence. The Kupiec (1995) Proportion-of-Failures test checks whether the empirical hit rate $\hat{\pi} = x/T$ matches the nominal τ through a likelihood ratio that is asymptotically χ^2_1. But frequency is only half the story: Kupiec is blind to timing. The Christoffersen (1998) combined test models the hit sequence as a first-order Markov chain and tests jointly whether breaches arrive at the right rate and at independent times:

$$LR_{CC} = LR_{UC} + LR_{\text{Ind}} \sim \chi^2_2. \tag{33}$$

This is the more demanding coverage criterion and the one used to validate the production model. Table 1 summarises the results. The corresponding coverage-convergence plot for the main models is shown in Figure **??**.

Table 1: Coverage backtest results at $\tau = 0.95$, $h = 5$. $\hat{\pi}$ is the realised share of days on which the return fell below the forecast (target 0.95). LR_{UC} is the Kupiec statistic; p_{UC} and p_{CC} are the p-values of the Kupiec unconditional and Christoffersen conditional coverage tests. All models fail to reject correct coverage at the 5% significance level.

Model	Test window	N	$\hat{\pi}$	LR_{UC} (p_{UC})	p_{CC}
Standard QReg	2023-01-02 – 2026-04-23	864	0.9387	2.189 (0.139)	0.306
CAViaR-indicator	2023-01-02 – 2026-04-20	517	0.9381	1.437 (0.231)	0.376
CAViaR-severity	2023-01-02 – 2026-04-20	517	0.9342	2.471 (0.116)	0.247

6.2 Model Selection and Forecast Horizons

The Diebold–Mariano test (Diebold and Mariano, 1995) arbitrates between the three models. Using the asymmetric tick loss and a HAC variance with bandwidth $h-1$, which matches the MA($h-1$) autocorrelation of h-step direct forecast errors, it tests the null of equal predictive accuracy. At $\tau = 0.95$, $h = 5$, Table 2 presents the results.

Table 2: Pairwise Diebold–Mariano tests, tick loss, $\tau = 0.95$, $h = 5$. No pairwise comparison is significant even at the 10% level.

Comparison	α	t-stat	p-value	Verdict
Standard vs CAViaR-i	$+0.00901$	$+0.909$	0.364	no sig. difference
Standard vs CAViaR-s	-0.09372	-1.081	0.280	no sig. difference
CAViaR-i vs CAViaR-s	-0.10273	-1.183	0.237	no sig. difference

On the headline predictive-accuracy criterion the three models are statistically indistinguishable: the Diebold–Mariano test finds no significant difference in tick loss at the operative horizon. The companion error metrics, reported in Table 3, place the average tick loss at 0.2380 (Standard), 0.2290 (CAViaR-indicator) and 0.3317 (CAViaR-severity). The indicator augmentation therefore attains a marginally lower average tick loss and root-mean-squared error than the standard model at $h = 5$, while the severity augmentation is the least accurate on both counts; none of these differences, however, is statistically significant.

Table 3: Out-of-sample error metrics at $\tau = 0.95$, $h = 5$.

Model	RMSE	MAPE	Avg. tick loss
Standard QReg	4.228	892.19	0.2380
CAViaR-indicator	4.039	901.86	0.2290
CAViaR-severity	4.594	901.25	0.3317

Because the overarching objective of this thesis is to forecast the full conditional distribution of Brent returns across a span of horizons, the standard quantile-regression direct forecast is retained as the baseline production model: it is parsimonious, passes every coverage backtest, and is statistically tied on tick loss. The CAViaR-indicator specification is carried forward as a tail-robust complement, since it delivers the lowest average tick loss and RMSE at the operative horizon and a comparable containment rate, the property that matters most when the model is used to price extreme geopolitical risk. The loss-minimising horizon for each model and quantile is reported in Table 4; the standard model attains its lowest upper-tail loss at the shortest horizons, whereas the CAViaR variants push their loss-minimising horizon outward, consistent with the fact that their breach regressors carry the sharpest signal when the horizon is short.

Table 4: Loss-minimising horizon per model and quantile (lower average pinball loss is better). Entries read $h^* \to$ loss.

Model	$\tau = 0.05$	$\tau = 0.50$	$\tau = 0.95$
Standard QReg	$h = 2 \to 0.3017$	$h = 4 \to 0.9375$	$h = 3 \to 0.2856$
CAViaR-indicator	$h = 9 \to 0.2960$	$h = 6 \to 0.9742$	$h = 7 \to 0.2806$
CAViaR-severity	$h = 20 \to 0.3397$	$h = 6 \to 0.9573$	$h = 16 \to 0.3810$

Table 5: Evidence summary across the three direct-forecasting models at $\tau = 0.95$, $h = 5$.

Test	What it certifies	Standard	CAViaR-i	CAViaR-s
Pinball / gen. gap	Proper-scoring fit, no overfit	Pass	Pass	Pass
Fallout audit	No catastrophic overshoot	97.83%	97.17%	97.26%
Kupiec POF	Correct breach frequency	Pass (0.139)	Pass (0.231)	Pass (0.116)
Christoffersen	Frequency + independence	Pass (0.306)	Pass (0.376)	Pass (0.247)
Diebold–Mariano	Relative accuracy	baseline	n.s.	n.s.

6.3 Backtesting the Risk Measures

Risk measures are only as credible as the evidence that they perform as advertised. A Value-at-Risk at the 97.5 percent confidence level is, implicitly, a promise: in no more than 2.5 percent of periods should the realised loss exceed the stated threshold. If exceedances occur materially more often, the model is understating risk and the capital reserves it implies are insufficient. If exceedances are materially rarer, the model is overly conservative and the capital it reserves is unnecessarily large, imposing an economic cost in the form of foregone investment.

The framework implements three progressively more stringent tests for VaR and two specialised tests for Expected Shortfall. The Kupiec unconditional coverage test counts the number of times the realised return exceeds the VaR threshold and tests whether this count is consistent with the nominal exceedance probability $\alpha = 1 -$ confidence:

$$LR_{UC} = -2\ln\left[\frac{\alpha^x(1-\alpha)^{n-x}}{\hat{\pi}^x(1-\hat{\pi})^{n-x}}\right], \quad \hat{\pi} = \frac{x}{n}. \tag{34}$$

The Christoffersen test augments this by also examining whether exceedances are independent over time. The Dynamic Quantile test of Engle and Manganelli (2004) is the most powerful of the three, regressing the hit sequence on lagged hits and the contemporaneous VaR forecast. The Basel Traffic Light system (Basel Committee on Banking Supervision, 2019) maps the number of exceedances observed over the most recent 250 trading days into Green, Yellow, or Red zones using the binomial distribution under the null of correct calibration.

For Expected Shortfall, the Acerbi and Székely (2014, 2017) tests exploit the joint elicitability of VaR and ES, constructing test statistics based on the severity of VaR exceedances rather than merely their frequency. The Costanzino and Curran (2015, 2018) ES Traffic Light test extends the Basel zone framework from VaR to ES, using a generalised coverage statistic calibrated under the FRTB parameters of $\alpha = 2.5\%$ and $N = 250$ observations.

Together, these backtesting procedures provide an evaluation of both the threshold accuracy (VaR) and the tail-severity accuracy (ES) of the *Oil-at-Risk* framework's risk measures. The complete derivation, hypotheses, and test statistics for every backtest are collected in Appendix C.

7 Early Warning System

7.1 Motivation: From Vulnerability Measurement to Anticipation

The risk and vulnerability metrics developed in the preceding section provide, at each forecast origin, a snapshot of the conditional distribution of future Brent returns. The Kullback–Leibler divergence, in particular, measures how far the conditional density has departed from its reference state, quantifying the degree to which the market's return distribution has moved away from the regime in which traditional risk metrics remain reliable. A natural follow-up question is whether this departure carries predictive content. If a spike in tail entropy today tends to precede large oil price movements in subsequent days, then the entropy measure may function not merely as a contemporaneous diagnostic but as an early warning indicator, one that could alert risk managers and policymakers to emerging vulnerabilities before they materialise in realised returns.

The distinction between a contemporaneous risk measure and a leading indicator matters in practice. A risk manager at a commodity trading desk who observes elevated left-tail entropy at the close of trading today faces two different operational realities depending on whether the signal is coincident or anticipatory. If the entropy spike merely reflects information already embedded in today's returns, it adds nothing that was not already available from the return series itself. If, however, the entropy spike tends to precede large return movements by several trading days, it provides a window during which hedging positions can be adjusted, margin requirements can be recalibrated, and contingency plans can be activated before the stress event arrives.

This section subjects the early warning hypothesis to two independent statistical tests. The analysis proceeds in three stages. First, the choice of the entropy indicator and its configuration are motivated on economic and statistical grounds. Second, the Cross-Correlation Function is used to identify the lead-lag structure between the entropy measures and Brent returns, examining whether the peak correlation occurs at a positive lag, which would suggest that the entropy series leads the target. Third, the Granger causality test is applied to assess whether lagged values of the entropy series carry statistically significant predictive information for future returns beyond the information already contained in the returns' own history.

7.2 Choice of the Early Warning Indicator

The entropy measures constructed in the preceding section offer several candidate indicators for the early warning system. Two design decisions must be made: the choice of the reference baseline against which the conditional density is compared, and the choice of the estimation window used to produce the entropy series.

Regarding the baseline, the framework provides two alternatives. The first is the unconditional Johnson S_U density fitted by Maximum Likelihood to the historical returns, which is the reference used in the original Vulnerable Growth framework of Adrian et al. (2019). The second is a Normal density calibrated to the historical mean and standard deviation of the return series. The unconditional JSU baseline captures genuine regime shifts in the conditional distribution, because the reference itself already accommodates the skewness and heavy tails that characterise oil returns. The Normal baseline, by contrast, measures the departure from Gaussianity, a broader concept that captures not only regime shifts but also the structural non-normality that is a permanent feature of commodity return distributions.

For the purpose of early warning, the Normal baseline appears to be the more informative choice. The historical distribution of Brent returns is approximately Gaussian in its central mass, a regularity that the Kolmogorov–Smirnov tests in Section 5.6 are consistent with. The tails, however, deviate substantially from normality during episodes of market stress. The KL divergence against a Normal reference therefore functions as a detector of tail fattening: when the conditional density closely resembles a Gaussian, the divergence is near zero; when the tails fatten in response to geopolitical escalation, speculative volatility, or demand-side shocks, the divergence rises. This spike-and-revert dynamic is the type of pattern that an early warning system should be designed to detect. The entropy series against the Normal and the unconditional baselines are displayed in Figures **??** and **??**.

The second design decision concerns the estimation window. The expanding window accumulates all historical data up to each forecast origin, which produces a smooth entropy series but dilutes the influence of recent observations as the sample grows. The rolling window, by contrast, uses only the most recent $W = 1{,}008$ observations (four trading years), producing a more responsive signal that adapts more quickly to regime changes. For early warning purposes, the rolling window is preferable on three grounds. First, it is computationally more efficient, with cost per iteration of order $O(W)$ rather than $O(t)$, which matters in a real-time monitoring context. Second, it is better suited to capturing regime transitions, because the influence of a departing regime decays as its observations leave the window rather than being permanently embedded in the expanding average.

Third, the rolling window is the standard convention in regulatory risk management, as codified in the Basel III internal-model framework (Basel Committee on Banking Supervision, 2019), and adopting it here ensures consistency with the VaR and Expected Shortfall computations that use the same window size.

The forecast horizon for the entropy computation is set at $h = 5$ trading days, matching the operative horizon of the direct forecasting and risk-metric steps. At shorter horizons such as $h = 1$ or $h = 2$, the quantile regression model tends to exhibit somewhat stronger predictive power, but a one- or two-day-ahead entropy signal provides too narrow a window for operational response. At longer horizons such as $h = 22$ (one trading month), the quantile regression coefficients become noisier and the MDE fitting inherits that noise, degrading the quality of the conditional density and, consequently, the reliability of the entropy measure. The five-day horizon occupies a middle ground where the conditional density retains meaningful predictive content while providing a sufficiently wide anticipation window for risk management action.

The early warning indicator is therefore defined as the rolling-window KL divergence of the conditional Johnson S_U density against the Normal baseline at $h = 5$, decomposed into three components: the full-distribution entropy, the left-tail entropy (capturing downside vulnerability), and the right-tail entropy (capturing upside vulnerability). The decomposition follows the convention established in Section 5.7, with the tail mark set at $\tau = 0.95$ so that the left-tail component captures the divergence below the 5th percentile and the right-tail component captures the divergence above the 95th percentile of the Normal reference.

7.3 The Cross-Correlation Function

The first diagnostic tool used to assess the anticipatory content of the entropy measures is the Cross-Correlation Function (CCF). The CCF is a standard instrument in the composite-indicator literature for identifying lead-lag relationships between economic time series (Bujosa et al., 2013), and its application to the early warning problem is direct: if the entropy series leads Brent returns, the CCF should exhibit its peak at a positive lag, indicating that today's entropy is most strongly correlated with future returns rather than with contemporaneous or past returns.

The CCF between two stationary time series $\{X_t\}$ and $\{Y_t\}$ at lag h is defined as the normalised cross-covariance:

$$r_{XY}(h) = \frac{c_{XY}(h)}{\sqrt{c_{XX}(0) \cdot c_{YY}(0)}}, \tag{35}$$

where $c_{XY}(h)$ is the sample cross-covariance at lag h. For non-negative lags ($h \geq 0$), the cross-covariance is computed as

$$c_{XY}(h) = \frac{1}{N} \sum_{t=1}^{N-h} (X_t - \bar{X})(Y_{t+h} - \bar{Y}), \tag{36}$$

and for negative lags ($h < 0$) the indices are reversed:

$$c_{XY}(h) = \frac{1}{N} \sum_{t=1-h}^{N} (X_t - \bar{X})(Y_{t+h} - \bar{Y}), \tag{37}$$

where $\bar{X}$ and $\bar{Y}$ are the sample means, and N is the number of observations (Brockwell and Davis, 1991). The denominator in Equation (35) normalises the cross-covariance by the product of the standard deviations $\sqrt{c_{XX}(0)}$ and $\sqrt{c_{YY}(0)}$, ensuring that the CCF is bounded in $[-1, 1]$ and directly interpretable as a correlation coefficient.

The sign convention adopted throughout is that $h > 0$ means the indicator X leads the target Y by h periods. This convention is standard in the composite-indicator literature (Bujosa et al., 2013) and has a clear economic interpretation: if the CCF between left-tail entropy and Brent returns peaks at $h^* = +16$, then today's left-tail entropy tends to be most strongly correlated with Brent returns 16 trading days hence.

Statistical significance of the CCF at each lag is assessed using Bartlett's approximation. Under the null hypothesis that the two series are independent, the sampling distribution of $r_{XY}(h)$ is approximately Normal with mean zero and standard deviation $1/\sqrt{N}$ for large N (Bartlett, 1946). The 95 percent confidence band is therefore $\pm 1.96/\sqrt{N}$, and any CCF value exceeding this band in absolute value is considered statistically significant at the 5 percent level. The estimated function is displayed in Figure **??**.

Three interpretive cases arise from the position of the peak correlation h^*. In the leading-indicator case ($h^* > 0$), the peak correlation occurs when the entropy series is aligned with future returns; this is the desired outcome for an early warning system, and the magnitude of h^* determines the anticipation horizon. In the coincident case ($h^* = 0$), the entropy series moves in lockstep with contemporaneous returns and has no predictive value. In the lagging case ($h^* < 0$), the indicator reacts to price movements after they have occurred and cannot serve as an early warning system.

Both the entropy indicators and the Brent return target are first-differenced before the CCF is computed, to ensure stationarity. Although the return series is stationary by construction, the entropy series can exhibit persistent level shifts during prolonged episodes of market stress, and differencing removes these non-stationary components while preserving the lead-lag dynamics of interest.

7.4 Granger Causality

The CCF identifies the lag at which the correlation between two series is maximised, but correlation does not establish predictive content in a rigorous econometric sense. Two series can be highly correlated at a positive lag because they are both driven by a common third factor, without the leading series carrying any incremental predictive information beyond what the target's own history already provides. The Granger causality test (Granger, 1969) addresses this limitation by asking a more precise question: do lagged values of the entropy indicator improve the forecast of Brent returns beyond the forecast that can be constructed from the returns' own lagged values alone?

The test is implemented by comparing two nested linear models. The restricted model regresses the target series on its own lagged values:

$$Y_t = \alpha_0 + \sum_{j=1}^{p} \alpha_j Y_{t-j} + e_t, \tag{38}$$

where Y_t denotes the first-differenced Brent return at date t, p is the lag order, α_0 is an intercept, and e_t is the error term. The unrestricted model augments this specification with lagged values of the entropy indicator:

$$Y_t = \alpha_0 + \sum_{j=1}^{p} \alpha_j Y_{t-j} + \sum_{j=1}^{p} \beta_j X_{t-j} + u_t, \tag{39}$$

where X_{t-j} denotes the first-differenced entropy indicator at lag j, and $\beta_1, \ldots, \beta_p$ are the coefficients on the lagged indicator values. The null hypothesis is that the indicator carries no incremental predictive content:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0. \tag{40}$$

Under this null, the lagged entropy values are jointly irrelevant for predicting future returns once the returns' own history is accounted for. Rejection of H_0 constitutes evidence that the entropy indicator Granger-causes Brent returns, in the specific sense that it carries information about future returns that is not already embedded in the returns' own autoregressive structure.

The test statistic is a standard F-test comparing the residual sum of squares of the restricted and unrestricted models:

$$F = \frac{(\mathrm{RSS}_r - \mathrm{RSS}_{ur})/p}{\mathrm{RSS}_{ur}/(T - 2p - 1)}, \tag{41}$$

where RSS_r and RSS_{ur} are the residual sums of squares of the restricted and unrestricted models respectively, p is the number of lags, and T is the effective sample size after accounting for the lag truncation. Under H_0, this statistic follows an $F(p, T - 2p - 1)$ distribution (Granger, 1969; Hamilton, 1994).

The lag order p is selected by the Akaike Information Criterion (AIC), which is the standard choice when the primary objective is predictive accuracy rather than model parsimony (Akaike, 1974). The AIC balances goodness of fit against model complexity by penalising each additional parameter:

$$\mathrm{AIC}(p) = \ln\left(\frac{\mathrm{RSS}_{ur}}{T}\right) + \frac{2k}{T}, \tag{42}$$

where $k = 2p + 1$ is the total number of estimated parameters in the unrestricted model. The criterion is evaluated for every candidate lag order from $p = 1$ to a maximum of $p_{\max} = 100$ trading days (approximately five calendar months), and the lag order that minimises the AIC is selected for the F-test. The wide search window allows the data to reveal lead-lag relationships that extend well beyond the short horizons typically examined in daily-frequency studies, at the cost of a larger parameter space that the AIC penalty is designed to discipline.

It is worth noting explicitly what Granger causality does and does not establish. The test provides evidence of incremental predictive content in a forecasting sense: lagged entropy values carry information about future returns that is not already captured by the returns' own lags. It does not establish structural or economic causation (Granger, 1969). The entropy indicator may lead returns because it captures the early stages of a distributional regime shift that subsequently manifests in realised price movements, or because both series respond to a common underlying driver with different speeds. The test is agnostic about the mechanism; it speaks only to the statistical predictability that an early warning system requires.

7.5 Internal Coherence of the Entropy Indicators

Before interpreting the lead-lag results, it is useful to examine whether the three entropy components, full, left-tail, and right-tail, move together or exhibit distinct dynamics. This is assessed by computing the pairwise CCF among all three indicators. If the components share a common factor, their pairwise CCFs should peak near zero lag with high correlation. If they respond to different types of shocks with different timing, the pairwise CCFs may reveal non-trivial lead-lag structure among the entropy measures themselves.

The coherence test applied here follows the internal-consistency diagnostic used in composite-indicator construction (Bujosa et al., 2013). In that framework, a well-constructed composite leading indicator should have components that are internally coherent, meaning they move together contemporaneously ($h^* \approx 0$) rather than leading or lagging one another. The expectation for the entropy indicators is similar but not identical: the full-distribution entropy is by construction a function of both tails, so it should be contemporaneously correlated with each tail component. The left-tail and right-tail entropies, however, respond to different types of shocks, downside and upside respectively, and need not be correlated with each other.

7.6 Results

The early warning battery is applied to the three rolling-window entropy indicators (full, left-tail, and right-tail KL divergence against the Normal baseline) computed at $h = 5$ over the out-of-sample period from June 2012 to April 2026, comprising 2,167 daily observations after first-differencing.

7.6.1 Cross-Correlation Function Results

The CCF analysis indicates that all three entropy indicators appear to lead Brent returns, with the peak correlation occurring at strictly positive lags in every case. The full-distribution entropy exhibits its peak cross-correlation at $h^* = +15$ trading days with $r = 0.311$, above the Bartlett 95 percent confidence band of ± 0.042. This suggests that a spike in the overall divergence of the conditional density from the Normal reference today tends to be associated with return movements approximately three trading weeks later. The right-tail entropy peaks at $h^* = +7$ with $r = 0.285$, suggesting that upside vulnerability signals may propagate to returns more quickly, within roughly one and a half trading weeks. The left-tail entropy peaks at $h^* = +16$ with $r = -0.149$; the negative sign is consistent with an inverse relationship between left-tail fattening (representing an accumulation of downside risk) and subsequent returns (which tend to decline when downside risk materialises), and the longer lead time of 16 trading days may reflect the more gradual build-up of downside risk as geopolitical tensions escalate or demand conditions deteriorate before the price impact is fully absorbed.

The observation that all three indicators exhibit $h^* > 0$ is the central finding of the CCF analysis: over this sample, the entropy measures behave as leading indicators rather than coincident or lagging diagnostics. The varying lead times across the three components may carry economic content. The right-tail entropy, which captures upside vulnerability, tends to translate to returns within about a week, which could be consistent with the speed at which supply-disruption premia are typically priced into the futures curve. The left-tail and full-distribution entropies, which capture broader distributional stress including the accumulation of downside vulnerability, appear to require two to three weeks to manifest in realised returns, which would be consistent with the generally slower propagation of demand-side shocks and the more gradual repricing of risk premia during periods of elevated macroeconomic uncertainty.

It should be noted, however, that the CCF is a bivariate measure and does not control for the influence of other variables. The observed lead-lag patterns could in principle be driven by a common third factor that affects the entropy series before it affects returns. The Granger causality test in the next subsection provides a partial control for this concern by conditioning on the returns' own history.

7.6.2 Granger Causality Results

The Granger causality tests provide additional support for the CCF findings. With the AIC search window extended to $p_{\max} = 100$ lags, the criterion selects lag orders ranging from 24 to 33 trading days, indicating that the predictive relationship, if genuine, extends over approximately one to one and a half calendar months.

Table 6: Granger causality test results: entropy indicators → Brent returns.

Indicator	F-statistic	p-value	Selected lag (p)	Significance
Full Entropy	22.222	< 0.001	33	***
Left Entropy	4.073	< 0.001	24	***
Right Entropy	24.364	< 0.001	31	***

*** significant at the 1% level. Lag order selected by AIC with $p_{\max} = 100$.
Sample: 2,167 daily observations (June 2012 – April 2026), first-differenced.
Source: Granger (1969); lag selection via Akaike (1974).

All three entropy indicators appear to Granger-cause Brent returns at the 1 percent significance level. The null hypothesis that the lagged entropy values carry no incremental predictive content beyond the returns' own autoregressive structure is rejected in every case, with p-values below 10^{-3}.

The F-statistics suggest an ordering of predictive content across the indicators. The right-tail entropy exhibits the largest F-statistic ($F = 24.364$, $p = 31$ lags), followed by the full-distribution entropy ($F = 22.222$, $p = 33$ lags) and the left-tail entropy ($F = 4.073$, $p = 24$ lags). One possible interpretation is that the upside tail of the return distribution in crude oil markets is disproportionately driven by supply-side shocks, geopolitical escalation, and speculative positioning, phenomena that tend to be abrupt and to generate directional price movements. The conditional density may detect the onset of these episodes through the fattening of the right tail, and the Granger test suggests that this signal carries information about future returns not already contained in the returns' own history.

The left-tail entropy, while still statistically significant, exhibits a considerably lower F-statistic. This could be consistent with the nature of downside risk in oil markets: demand-side contractions tend to unfold more gradually than supply disruptions, and the return series itself, through its own autoregressive structure, may already absorb much of the predictable component of these slow-moving declines. The incremental information contributed by the left-tail entropy would therefore be smaller in magnitude.

A caveat is in order regarding the interpretation of these results. The relatively high lag orders selected by the AIC imply a large number of parameters in the unrestricted model (67 parameters for $p = 33$, for example). While the sample size of 2,167 observations provides reasonable degrees of freedom, the possibility of overfitting or spurious in-sample predictability cannot be entirely excluded without a formal out-of-sample forecasting exercise. The Granger test results should therefore be read as suggestive evidence of predictive content rather than as a definitive statement of forecasting ability.

7.6.3 Internal Coherence

The pairwise CCF analysis among the three entropy indicators reveals a pattern of partial coherence, reported in Table 7.

Table 7: Internal coherence: pairwise CCF among entropy indicators.

Series X	Series Y	h^*	r at h^*	Significant
Full Entropy	Left Entropy	0	+0.748	Yes
Full Entropy	Right Entropy	0	+0.631	Yes
Left Entropy	Right Entropy	0	−0.036	No

Bartlett 95% confidence band: $\pm 1.96/\sqrt{N}$.

The full-distribution entropy is strongly and contemporaneously correlated with both the left-tail component ($r = 0.748$, $h^* = 0$) and the right-tail component ($r = 0.631$, $h^* = 0$). This is expected by construction, since the full entropy integrates over the entire support of the density, including both tails. The left-tail and right-tail entropies, however, are essentially uncorrelated ($r = -0.036$, not significant), which is consistent with the interpretation that they respond to distinct types of market events. A supply-disruption shock that fattens the right tail of the return distribution does not necessarily fatten the left tail simultaneously, and a demand-contraction episode that elevates left-tail entropy need not generate upside vulnerability. If this interpretation is correct, the two tail components carry largely independent information, which would explain why monitoring both separately, rather than relying on the full-distribution aggregate alone, may provide a richer early warning profile.

7.7 Synthesis and Implications

The combined evidence from the Cross-Correlation Function and the Granger causality test is consistent with the hypothesis that the entropy measures derived from the *Oil-at-Risk* framework contain early warning information about oil price movements over the sample studied. The main observations are as follows. All three entropy indicators exhibit peak cross-correlations with Brent returns at positive lags, ranging from 7 to 16 trading days, so that over this sample they behave as leading rather than contemporaneous diagnostics, with lead times that appear operationally meaningful. The Granger causality tests indicate that the predictive content of the entropy indicators is incremental: it is not replicable by the returns' own autoregressive structure alone. The right-tail entropy tends to provide the strongest and fastest signal, consistent with the abrupt, event-driven nature of supply-side shocks in crude oil markets, while the left-tail entropy provides a weaker but still statistically significant signal, consistent with the slower propagation of demand-side vulnerabilities. Finally, the internal coherence analysis indicates that the left-tail and right-tail entropies respond to largely distinct phenomena and carry independent information about different types of oil market risk.

From a practical perspective, the system implies the following operational logic over the period examined. A risk manager monitoring the *Oil-at-Risk* dashboard observes the daily rolling entropy series decomposed by tail. A significant spike in right-tail entropy signals that the conditional distribution is developing an unusually fat right tail, potentially driven by geopolitical escalation or supply tightening, and that this stress may manifest in realised price movements within approximately one to two trading weeks. A significant spike in left-tail entropy signals the accumulation of downside vulnerability, potentially associated with weakening demand conditions or an appreciating dollar, with a somewhat longer anticipation window. The full-distribution entropy, which aggregates both tails, provides the broadest signal. These lead times are sufficiently long to permit meaningful risk management responses: adjusting hedge ratios, recalibrating margin requirements, or revising scenario assumptions for stress testing. The entropy-based indicator therefore extends the distributional analysis of the *Oil-at-Risk* framework from a retrospective diagnostic toward a forward-looking surveillance tool.

8 Conclusion

This thesis has developed, estimated, and evaluated a probabilistic framework for assessing tail-risk vulnerabilities in crude oil prices under geopolitical uncertainty. The framework, designated *Oil-at-Risk*, adapts the *Growth-at-Risk* paradigm of Adrian et al. (2019) from macrofinancial surveillance to the energy commodity domain, and enriches it along several dimensions that address the specific statistical and economic challenges posed by daily Brent crude oil returns.

The empirical pipeline begins with quantile regression of Brent log-returns on a parsimonious set of geopolitical and macroeconomic conditioning variables, estimated via direct multi-step forecasting to avoid the compounding bias of iterated schemes. Three specifications were fielded: a standard quantile regression, a CAViaR-indicator augmentation that introduces binary breach memory, and a CAViaR-severity augmentation that encodes the magnitude of past exceedances. The direct forecasting design produced a genuine out-of-sample evaluation over more than three years of daily data at the operative horizon $h = 5$, with a rolling window of 1,008 observations and a retraining cadence of 30 days. The Wald test for coefficient equality across quantiles supported the identifying premise: geopolitical risk exerts heterogeneous effects across the conditional distribution, compressing the lower tail more aggressively than the upper, a finding consistent with a quantile-based approach rather than a conditional mean.

The discrete quantile skeleton was interpolated into a continuous conditional density using the Johnson S_U distribution, fitted by Minimum Distance Estimation at each forecast origin. The S_U family was chosen over the Azzalini skewed-t used in the original *Growth-at-Risk* framework for its independent control over skewness and kurtosis and its closed-form quantile function, which removes the numerical root-finding that the skewed-t requires. The Probability Integral Transform evaluated on out-of-sample data indicated that the fitted densities provide a statistically adequate characterisation of the conditional distribution: the Kolmogorov–Smirnov test failed to reject uniformity of the PIT sequence.

From the fitted conditional densities, four complementary risk metrics were extracted at every forecast origin: Value-at-Risk, Expected Shortfall, Kullback–Leibler divergence (decomposed into left-tail, centre, and right-tail components), and distributional skewness. These metrics were computed at the 97.5 percent confidence level used under Basel III and benchmarked against unconditional historical simulation. The conditional and unconditional VaR series tracked each other closely during tranquil periods but diverged during geopolitical escalations, indicating that the conditioning variables detect regime shifts that the backward-looking historical measure misses. The KL divergence, computed against both a Normal and an unconditional S_U baseline, provided a global measure of distributional departure that captures the wholesale reshaping of the conditional density, not merely the movement of a single quantile threshold.

The backtesting battery subjected the risk measures to a set of progressively more stringent tests. The Kupiec unconditional coverage test, the Christoffersen conditional coverage test, and the Engle–Manganelli dynamic quantile test were applied uniformly across all model variants at the operative horizon $h = 5$. All three specifications passed every test at the 5 percent significance level: breach frequencies were correct, breach sequences were serially independent, and the realised return remained inside the predicted envelope on roughly 97 to 98 percent of test days. The Diebold–Mariano test found no statistically significant difference in tick loss among the three models at the operative horizon. The standard quantile regression was retained as the production model on grounds of parsimony and its passing of every backtest, while the CAViaR-indicator specification was carried forward as a tail-robust complement, delivering the lowest average tick loss and root-mean-squared error at the operative horizon.

The entropy-based early warning indicator was constructed as the rolling-window Kullback–Leibler divergence of the conditional density against a Normal baseline, decomposed by tail, and its anticipatory content was examined with the Cross-Correlation Function and the Granger causality test. Over the sample studied, all three entropy components peaked at positive lags of 7 to 16 trading days and Granger-caused Brent returns at the 1 percent level, with the right-tail component carrying the strongest and fastest signal. These results are read as suggestive rather than definitive, given the bivariate nature of the CCF and the high dimensionality of the Granger models, but they indicate that the distributional output of the *Oil-at-Risk* framework carries information beyond a contemporaneous snapshot.

Taken together, these results deliver the contributions set out in Section 3. The *Oil-at-Risk* concept has been operationalised and evaluated out of sample, showing that the *Growth-at-Risk* paradigm transfers to energy commodity markets. The CAViaR augmentation has been implemented within the direct-forecasting design and compared against the standard specification. The Johnson S_U distribution has been used to track the time-varying asymmetry of oil return distributions across geopolitical regimes, offering a computationally convenient alternative to the skewed-t in distributional *Growth-at-Risk* applications. The entropy-based early warning indicator has been defined and tested for lead-lag content. The backtesting battery has been applied uniformly across all specifications. Finally, the open-source Python module that implements the full pipeline is intended to be publicly available in the author's repository, enabling replication and extension.

The framework operates under clear boundaries. The direct forecasting approach, while immune to the compounding bias of iterated schemes, does not exploit the parametric efficiency that a correctly specified dynamic model would provide. The Johnson S_U imposes a unimodal density, which cannot represent the bimodality that may arise when the market is unable to resolve the direction of a geopolitical shock. The retraining cadence of 30 days introduces a modest staleness in the regression coefficients between refit dates, although the robustness check with daily retraining indicates that this has negligible impact on the risk metrics. These boundaries delimit the scope of the conclusions drawn here, all of which rest on the evidence assembled across the in-sample diagnostics, the out-of-sample backtests, and the lead-lag analysis of the entropy indicator.

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A Optimisation Method and Hypotheses

In standard OLS, because the squared loss function creates a smooth parabola, taking the derivative and setting it to zero yields a closed-form algebraic solution: $\hat{\beta} = (X^\top X)^{-1} X^\top Y$. However, the absolute value in the check function creates a sharp "V" shape at zero. This non-differentiable corner means that standard first-order conditions cannot be applied. Instead, the optimisation is transformed into a Linear Programming (LP) problem.

To eliminate the absolute piecewise nature of the function, every residual is split into two strictly non-negative directional buckets: u_i^+ (the under-prediction bucket, measuring the magnitude by which the actual value exceeded the prediction) and u_i^- (the over-prediction bucket, measuring the magnitude by which the prediction exceeded the actual value). Every residual can now be expressed as a linear combination $u_i = u_i^+ - u_i^-$. Substituting this into the regression equation ($y_i - x_i^\top \beta = u_i^+ - u_i^-$) and rearranging defines the structural constraints for the optimisation algorithm:

$$x_i^\top \beta + u_i^+ - u_i^- = y_i. \tag{43}$$

Applying this transformation removes the absolute value operator from the objective function. The minimisation problem is expressed in a standard primal Linear Programming format:

$$\min_{\beta,\, u^+,\, u^-} \sum_{i=1}^{n} \Big[\tau\, u_i^+ + (1-\tau)\, u_i^- \Big] \quad \text{subject to} \quad x_i^\top \beta + u_i^+ - u_i^- = y_i, \quad u_i^+,\, u_i^- \geq 0. \tag{44}$$

By formulating the problem this way, statistical software can solve the regression using efficient LP algorithms, primarily the Barrodale–Roberts modified simplex algorithm (Barrodale and Roberts, 1973) for smaller datasets, which searches for the minimum along the geometric edges of the data intersections, or Frisch–Newton interior-point methods (Portnoy and Koenker, 1997) for large datasets, which cut through the feasible space using a logarithmic barrier to ensure positivity.

For $\hat{\beta}_\tau$ to be a consistent estimator, the model relies on several core hypotheses. Linearity of conditional quantiles requires that the conditional quantile has a linear relationship with the regressors, such that $Q_\tau(Y \mid X) = X^\top \beta(\tau)$. Strict exogeneity requires that the conditional quantile of the error term, given X, is zero: $Q_\tau(u \mid X) = 0$. No perfect multicollinearity requires that the matrix $\mathbb{E}[XX^\top]$ is finite, positive-definite, and invertible. Positive density requires that the conditional density function of the error term around zero is strictly positive and finite ($0 < f_{u|X}(0 \mid X) < \infty$). Because precision in quantile regression depends on the density of observations near the estimated quantile, this condition ensures the estimator is well-identified.

B Quantile Regression Diagnostics: Full Technical Development

This appendix presents the methodological foundation of each diagnostic test applied to the specification. For each test, the statistical framework, null hypothesis, test statistic, and asymptotic distribution are stated.

B.1 Goodness-of-Fit: Koenker–Machado $R^1(\tau)$

The local goodness-of-fit measure introduced by Koenker and Machado (1999) defines a pseudo-R^2 as the proportional reduction in the asymmetrically weighted sum of absolute errors relative to an intercept-only null model:

$$R^1(\tau) = 1 - \frac{\sum_{t=1}^{T} \rho_\tau\Big(y_t - x_t'\hat{\beta}(\tau)\Big)}{\sum_{t=1}^{T} \rho_\tau\Big(y_t - \hat{\xi}_\tau\Big)}, \tag{45}$$

where $\hat{\xi}_\tau$ is the unconditional sample τ-quantile. The metric is bounded in $[0, 1]$ and interpreted locally: a value of $R^1(\tau) = 0.10$ at $\tau = 0.05$ indicates that the regressors reduce the weighted objective by ten percent relative to the unconditional quantile at that tail. Koenker and Machado (1999) establish that the statistic $T \cdot V(\tau) \cdot R^1(\tau)$ is asymptotically chi-squared with k degrees of freedom.

B.2 Wald Test for Equality of Coefficients across Quantiles

The null hypothesis reads $H_0\colon R\zeta = r$, where ζ is the stacked coefficient vector and R is a restriction matrix imposing equality across quantile pairs. The Wald statistic is:

$$T_n = n\,(R\hat{\zeta} - r)^\top \Big[R\,\hat{V}_n\,R^\top\Big]^{-1} (R\hat{\zeta} - r) \xrightarrow{d} \chi^2_q, \tag{46}$$

where $\hat{V}_n$ is the joint asymptotic covariance matrix of $\hat{\zeta}$, constructed via a moving block bootstrap to account for temporal dependence.

B.3 Quantile Crossing Inspection

For any two quantile levels $\tau_1 < \tau_2$ and any observation t, the monotonicity condition $\hat{q}_t(\tau_1) \leq \hat{q}_t(\tau_2)$ must hold. Violations arise when the estimated linear quantile planes intersect in the covariate space, typically as a consequence of over-parameterisation, collinear regressors, or an insufficient number of observations. The proportion of crossing observations and the magnitude of the maximal inversion are reported as informal diagnostics.

B.4 Stationarity: Quantile Autoregressive Unit Root Test

Koenker and Xiao (2004) test for quantile-specific unit roots via the auxiliary error-correction regression:

$$\Delta y_t = \mu(\tau) + \rho(\tau)\, y_{t-1} + \sum_{j=1}^{p} \gamma_j(\tau)\, \Delta y_{t-j} + \epsilon_t(\tau). \tag{47}$$

The null $H_0\colon \rho(\tau) = 0$ posits a quantile-specific unit root. The test statistic $t_{\mathrm{QAR}}(\tau)$ does not follow a standard normal distribution under the null; quantile-specific critical values are obtained by simulation following Koenker and Xiao (2004).

B.5 Conditional Heteroskedasticity Test

The ARCH-LM test of Engle (1982), applied to the quantile residuals $\hat{\epsilon}_t(\tau) = y_t - x_t'\hat{\beta}(\tau)$, evaluates whether the squared residuals exhibit autoregressive conditional heteroskedasticity:

$$\hat{\epsilon}_t^2(\tau) = \alpha_0 + \sum_{i=1}^{p} \alpha_i\, \hat{\epsilon}_{t-i}^2(\tau) + \nu_t. \tag{48}$$

The Lagrange multiplier statistic $LM = T \cdot R^2$ is asymptotically χ_p^2 under $H_0\colon \alpha_1 = \cdots = \alpha_p = 0$. Failure to reject provides evidence that the estimated quantile coefficients are not contaminated by residual volatility clustering.

C Out-of-Sample Forecast Evaluation Tests

This appendix documents every test used to evaluate the direct-forecasting models. Two conventions hold throughout. First, a hit (or breach) is defined under a unified directional rule: a hit occurs whenever the realised value falls strictly below the forecast, $y_{t+h} < \hat{y}_{t+h|t}(\tau)$, so that for any quantile the model should be breached exactly $\tau \times 100\%$ of the time. Second, all coverage tests are run out-of-sample on a fixed 1,008-day rolling window, re-estimated at every forecast origin.

C.1 The Rolling-Window Experiment

Every statistic is computed on genuinely out-of-sample forecasts generated by a fixed-width rolling window. For each forecast origin t in the test span the model is trained on the 1,008 observations in the half-open interval $[t - 1008,\, t)$, a single h-step-ahead prediction of Y_{t+h} is produced, and the window rolls forward one day. Because the realised target Y_{t+h} never enters the training window, there is no load-bearing look-ahead. The reported figures at $h = 5$, $\tau = 0.95$ are averages over hundreds of origins: 864 out-of-sample forecasts for the benchmark (2 January 2023 to 23 April 2026) and 517 for the CAViaR variants (2 January 2023 to 20 April 2026), the difference reflecting the extra rows consumed by constructing the breach regressors.

C.2 Pinball Loss and Scoring

The asymmetric pinball (check) loss is the unique (up to affine transformation) strictly proper scoring rule for a quantile (Gneiting and Raftery, 2007):

$$L_\tau(e) = \begin{cases} \tau\, e & \text{if } e \geq 0, \\ (\tau - 1)\, e & \text{if } e < 0, \end{cases} \qquad e = y_{t+h} - \hat{y}(\tau). \tag{49}$$

If model A has a lower out-of-sample pinball loss than model B at quantile τ, then A is the better estimator of that τ-quantile; the comparison needs no distributional assumption.

C.3 Kupiec Proportion-of-Failures Test

With T out-of-sample forecasts and x observed hits at empirical rate $\hat{\pi} = x/T$ (Kupiec, 1995):

$$LR_{UC} = -2 \ln\left[\frac{(1-\tau)^{T-x}\, \tau^x}{(1-\hat{\pi})^{T-x}\, \hat{\pi}^x}\right] \sim \chi^2_1. \tag{50}$$

C.4 Christoffersen Independence Test

The hit sequence is modelled as a first-order Markov chain with transition probabilities π_{01} and π_{11} (Christoffersen, 1998):

$$LR_{\text{Ind}} = -2\ln\left(\frac{\mathcal{L}_{\text{indep}}}{\mathcal{L}_{\text{dep}}}\right) \sim \chi_1^2. \tag{51}$$

C.5 Christoffersen Combined Conditional-Coverage Test

$$LR_{CC} = LR_{UC} + LR_{\text{Ind}} \sim \chi_2^2. \tag{52}$$

A model passes only if its breaches arrive at the right frequency and at independent times.

C.6 Diebold–Mariano Test of Equal Predictive Accuracy

Let $d_t = L_{1,t} - L_{2,t}$ be the differential tick loss (Diebold and Mariano, 1995). The statistic standardises the mean differential by a HAC long-run variance:

$$DM = \frac{\bar{d}}{\sqrt{LRV/P}}, \quad LRV = \gamma_0 + 2\sum_{k=1}^{h-1}\gamma_k \sim t_{P-1}. \tag{53}$$

The HAC bandwidth is set to $h-1$, matching the MA($h-1$) autocorrelation of h-step direct forecast errors.

D Graphical Annex

The figures referenced throughout the text are collected here. Each placeholder is to be replaced by the corresponding image; the captions and cross-references are already in place. To insert an image, uncomment the `\includegraphics` line and remove the placeholder box.

Figure 1: Evolution of the European Brent crude oil spot price over the sample period. Source: U.S. Energy Information Administration (via FRED).

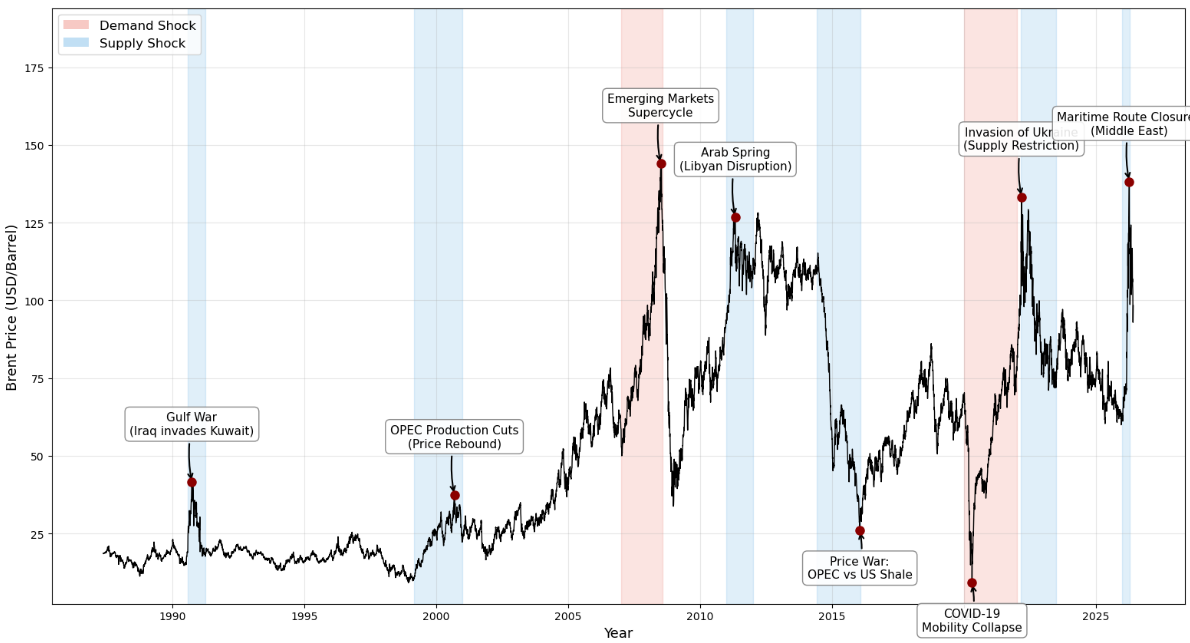

Figure 2: Enter Caption

Figure 3: In-sample quantile regression coefficient paths across τ for specification (5), with the OLS benchmark for comparison. Source: author's estimation.

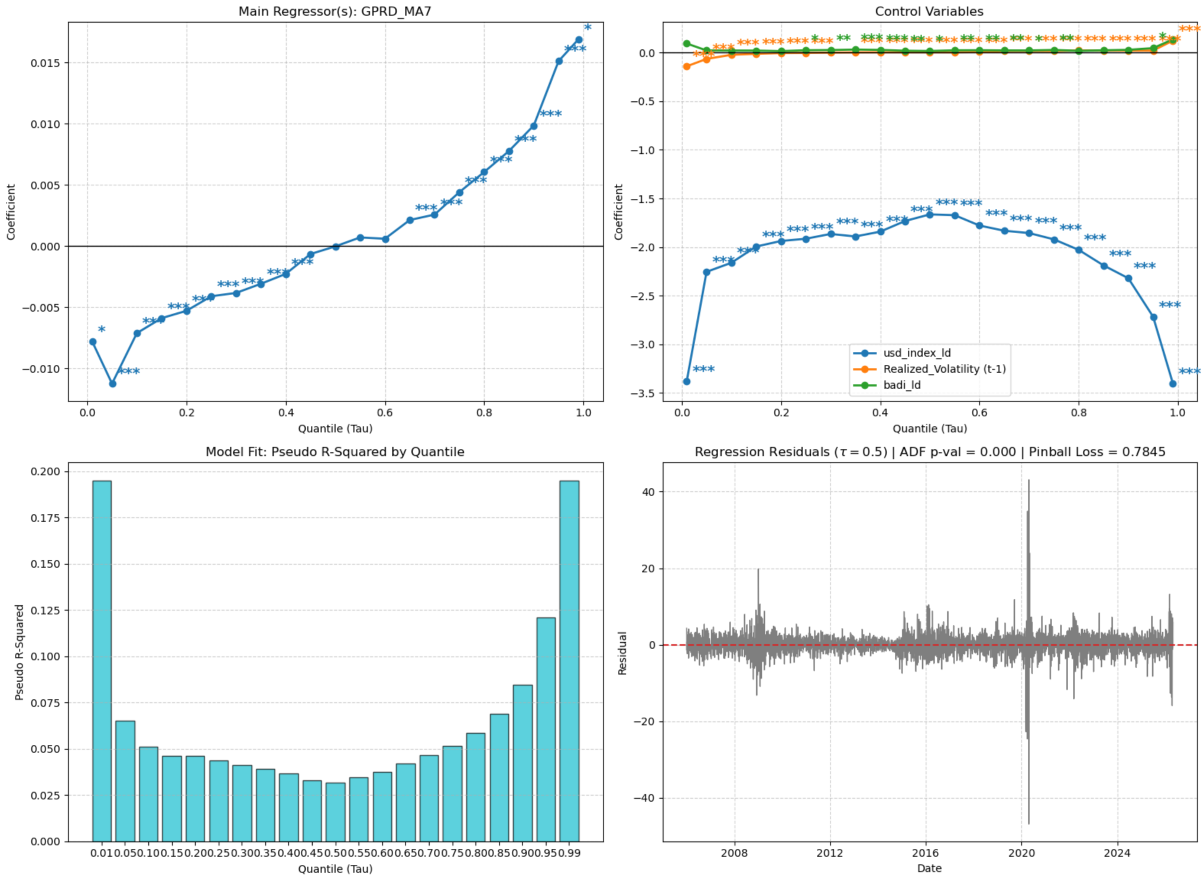

Figure 4: Three-dimensional representation of the in-sample fitted Johnson S_U conditional density across forecast origins. Source: author's estimation.

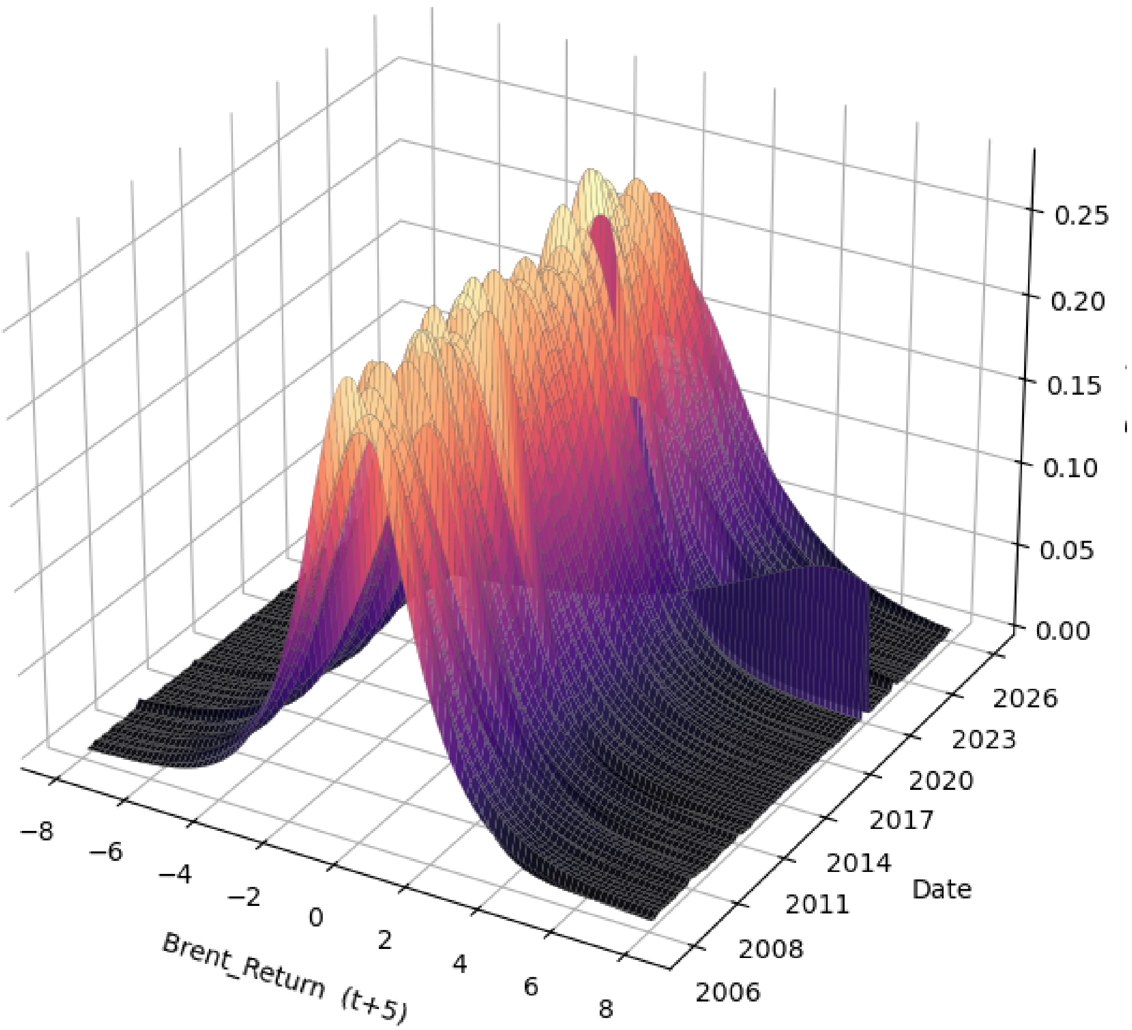

Figure 5: Realised Brent returns against the forecasted conditional quantile envelope, out-of-sample, $h = 5$. Source: author's estimation.

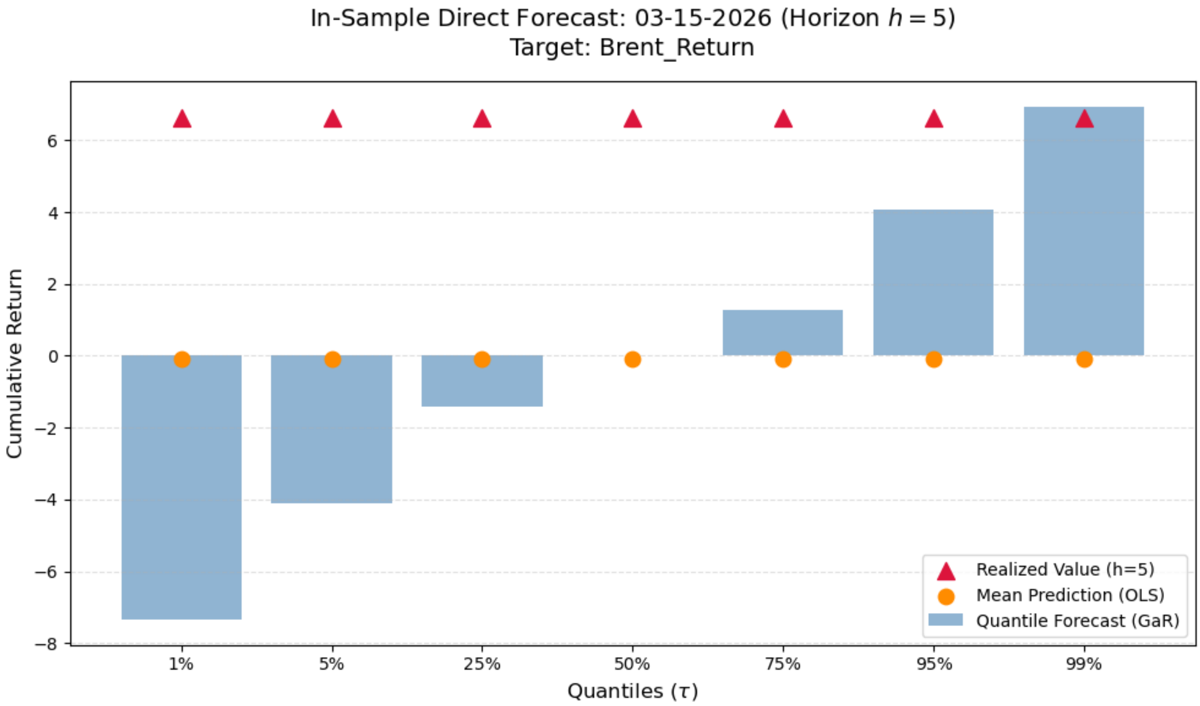

Figure 6: Enter Caption

Figure 7: Out-of-sample pinball loss across horizons for the three models, with the train–test generalisation gap. Source: author's estimation.

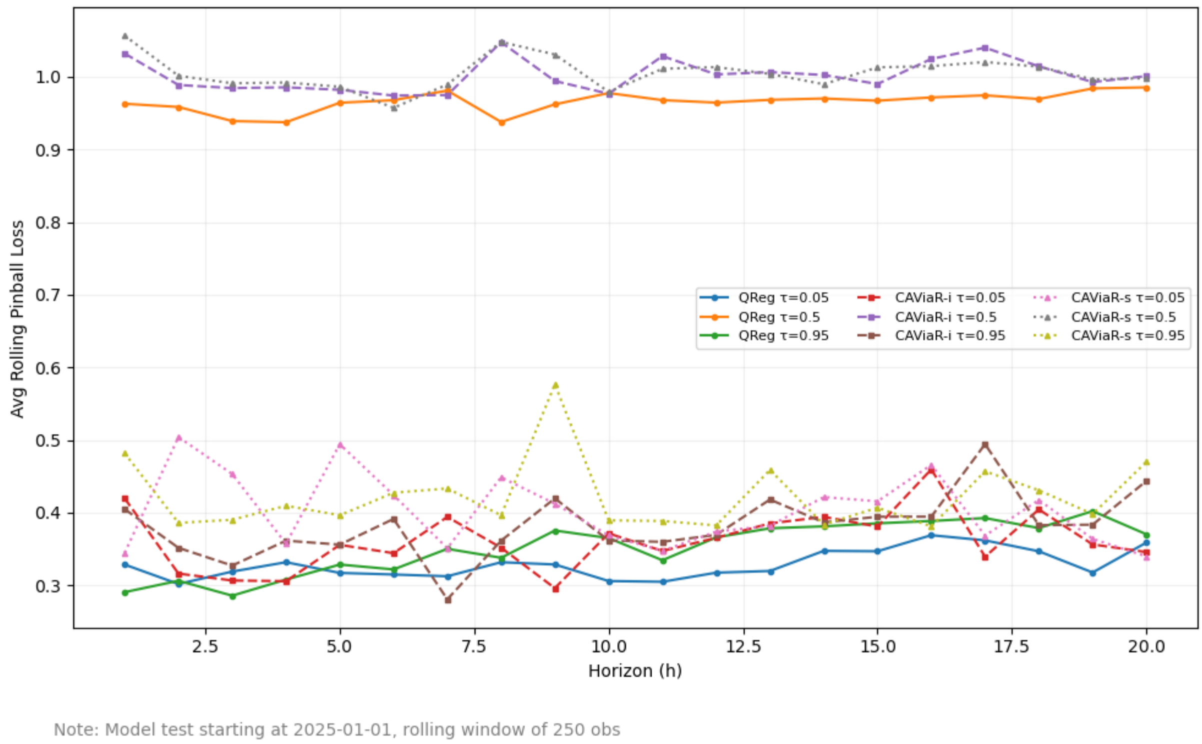

Figure 8: Enter Caption

Figure 9: Breach paths and fallout severity for the Standard model. Source: author's estimation.

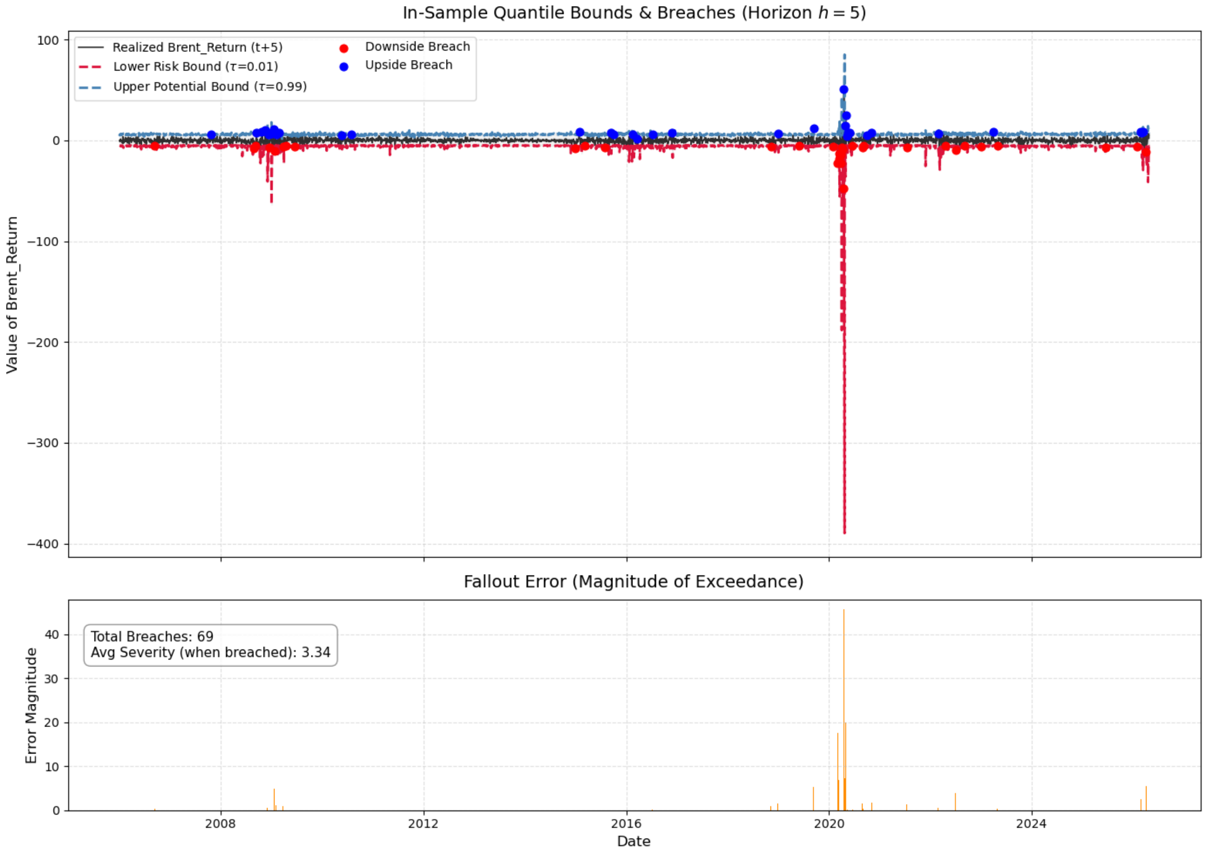

Figure 10: Breach paths and fallout severity for the CAViAR Indicator model. Source: author's estimation.

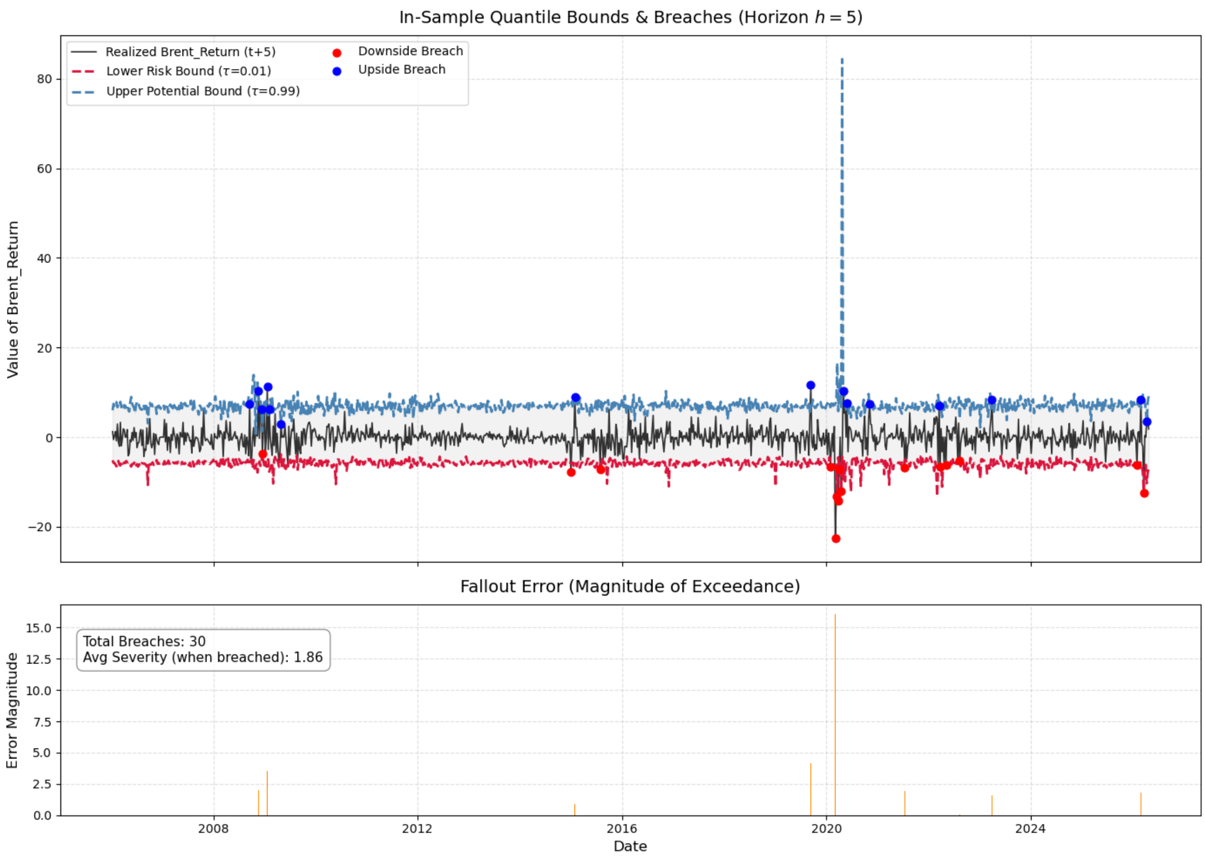

Figure 11: Breach paths and fallout severity for the CAViaR Fallout Error. Source: author's estimation.

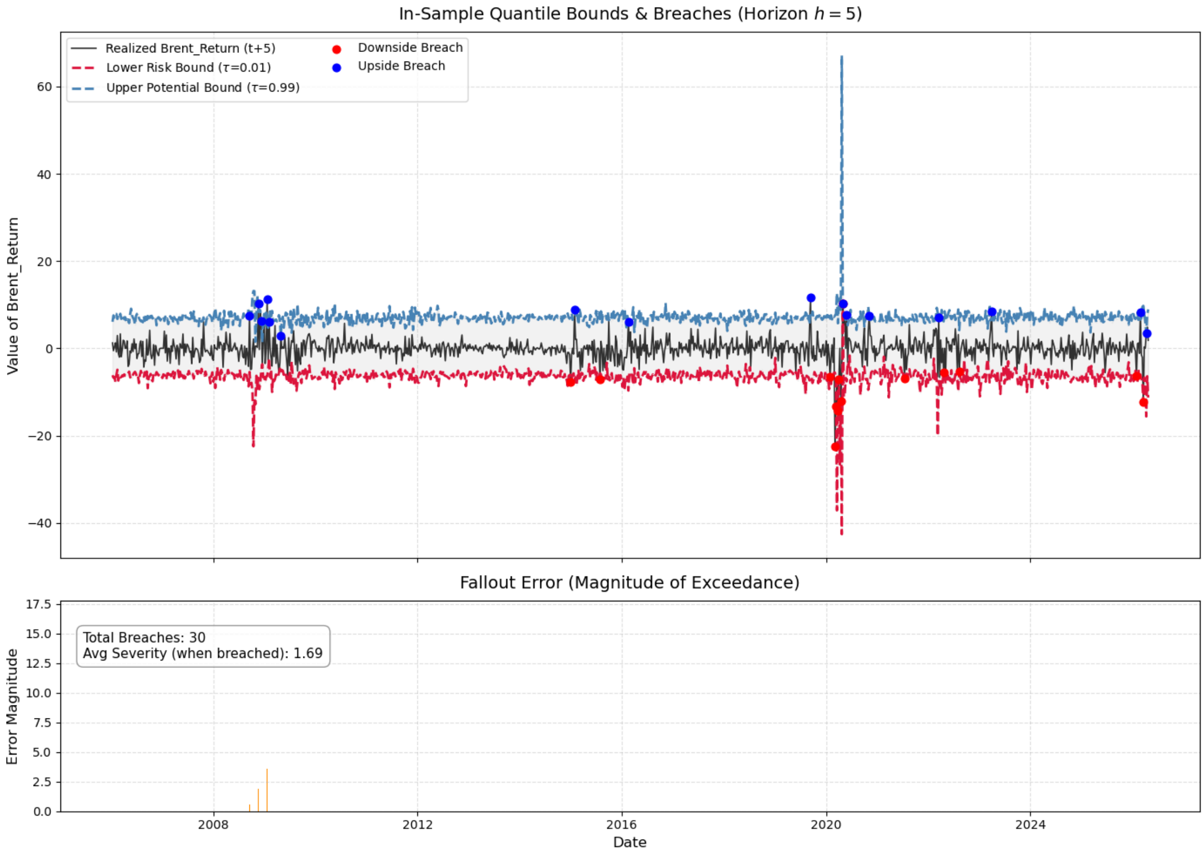

Figure 12: Coverage-convergence plot at $\tau = 0.95$, $h = 5$, for the main models. Source: author's estimation.

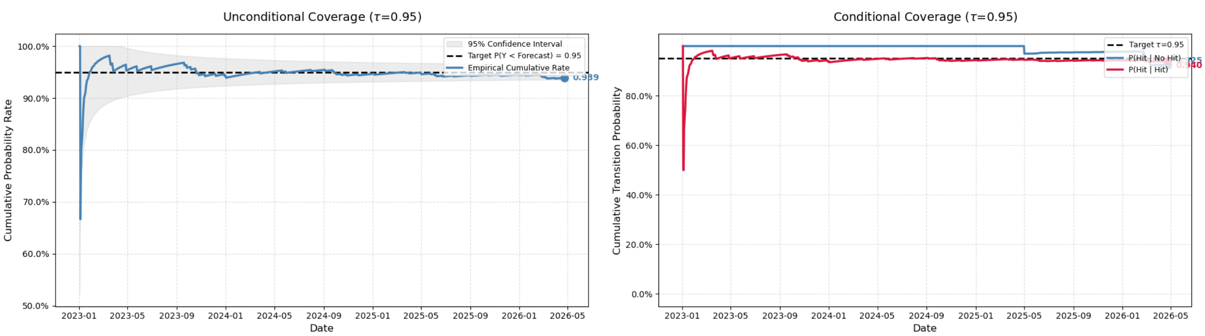

Figure 13: Unconditional Johnson S_U density fitted by Maximum Likelihood to historical Brent returns, used as a KL baseline. Source: author's estimation.

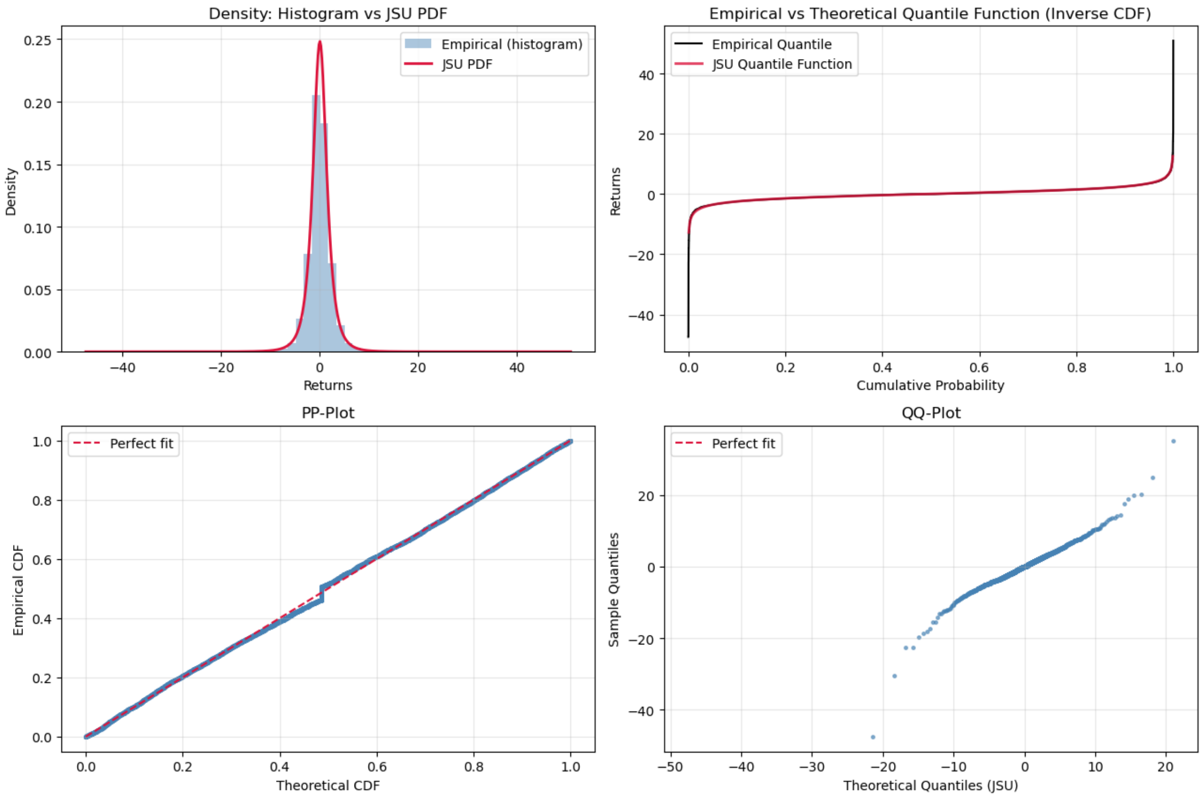

Figure 14: Probability Integral Transform histogram for the Johnson S_U conditional densities, $h = 5$, with the Kolmogorov–Smirnov statistic and p-value. Source: author's estimation.

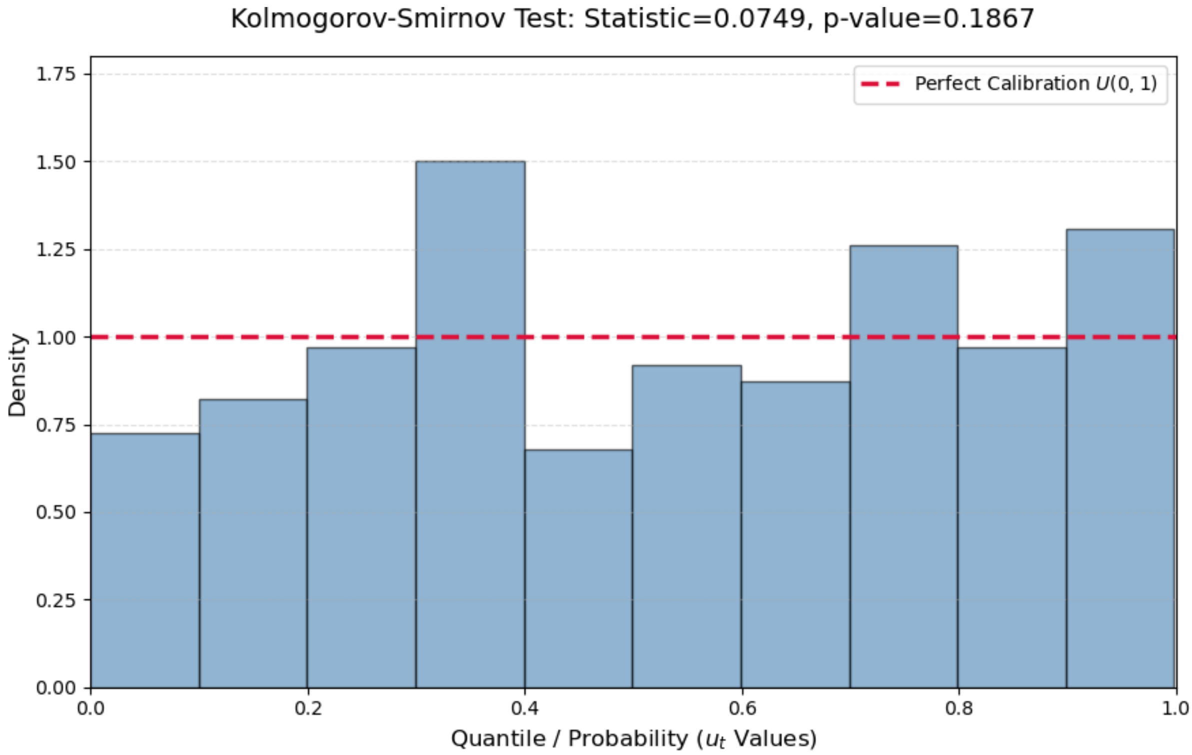

Figure 15: Probability Integral Transform diagonal plot (empirical vs. theoretical CDF). Source: author's estimation.

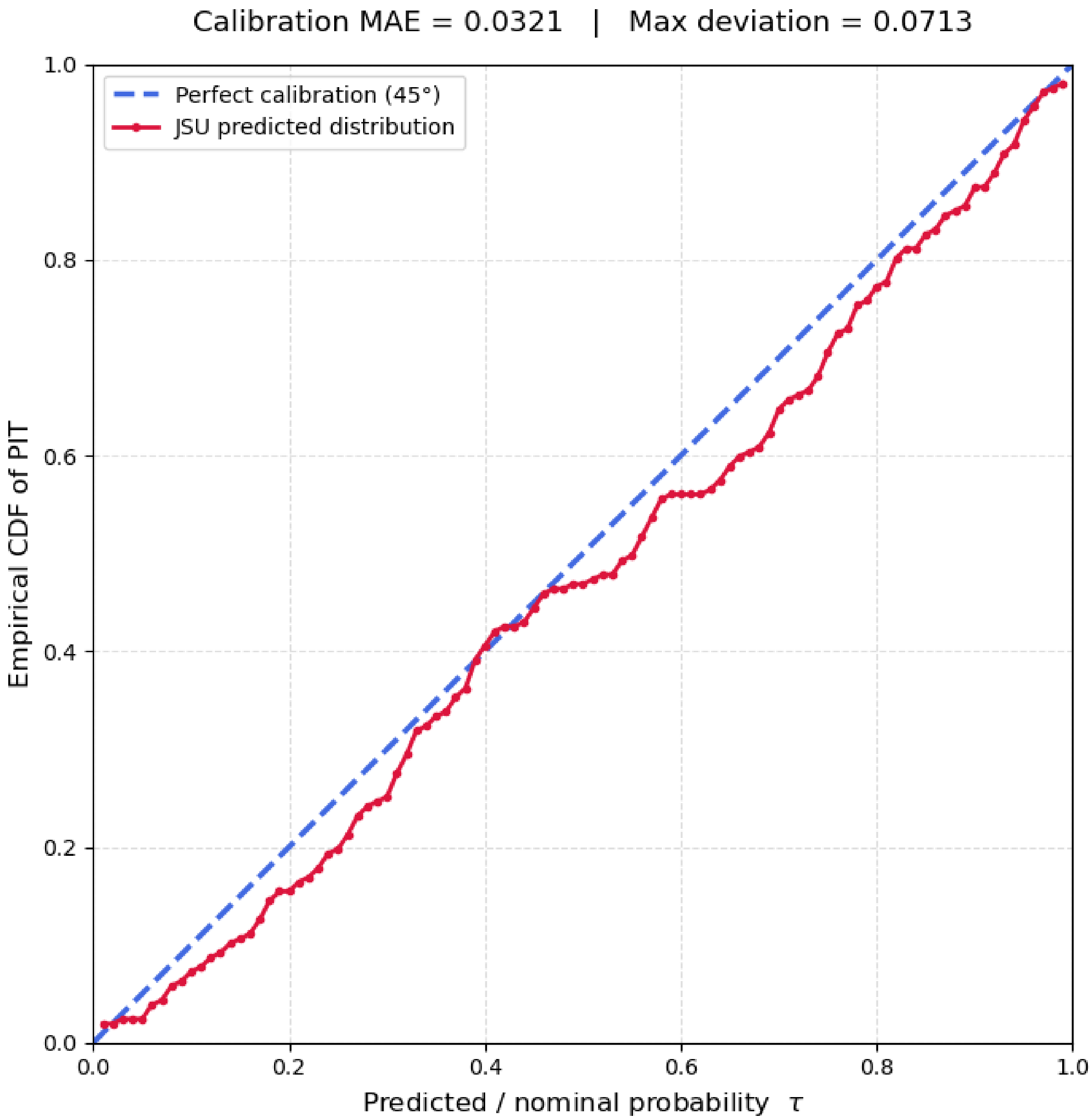

Figure 16: Kullback–Leibler entropy of the conditional density against the Normal baseline, expanding window. Source: author's estimation.

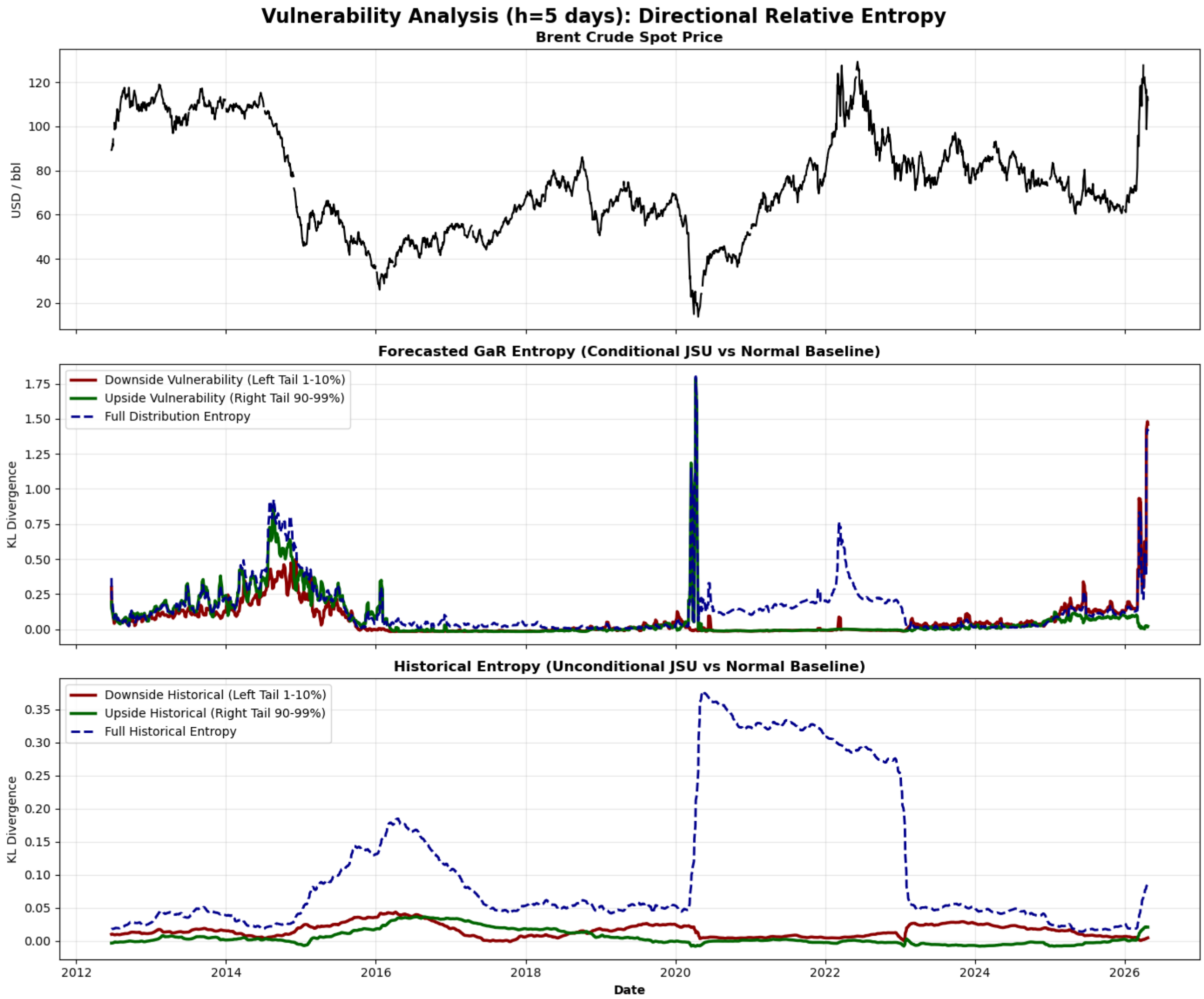

Figure 17: Kullback–Leibler entropy of the conditional density against the unconditional Johnson S_U baseline, expanding window. Source: author's estimation.

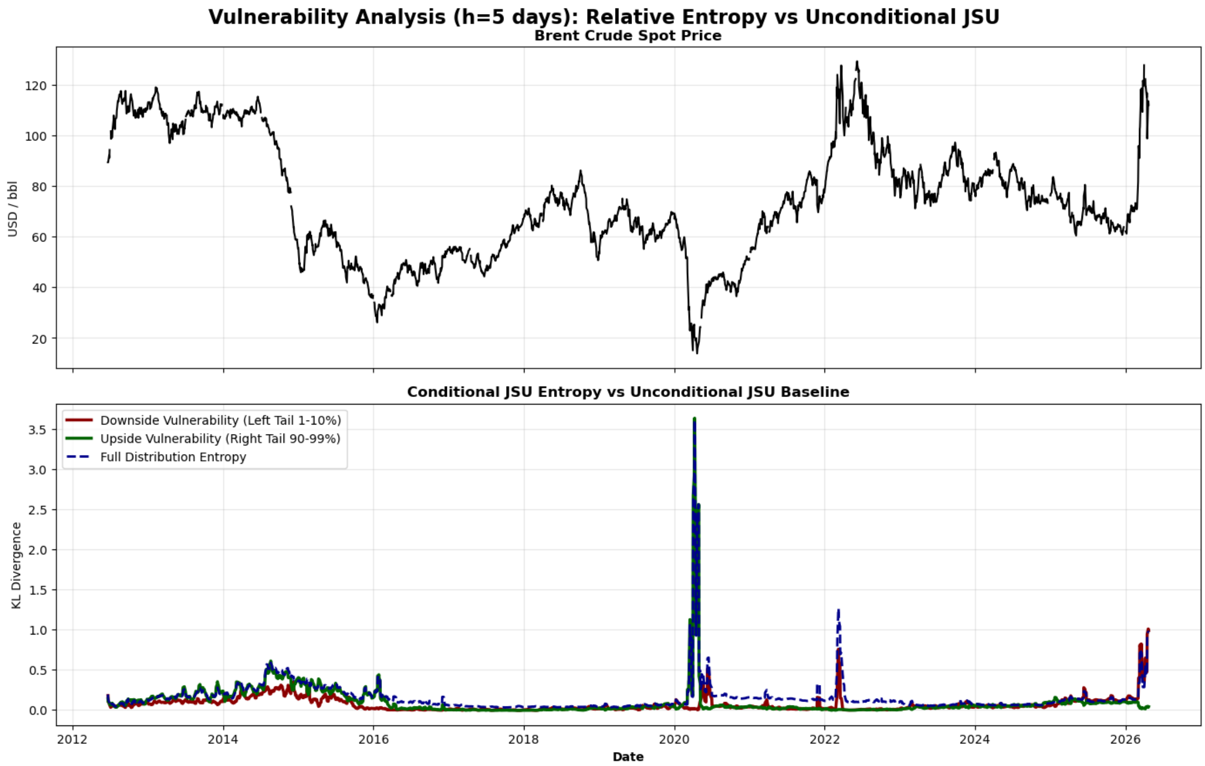

Figure 18: Conditional and unconditional Value -at-Risk (VaR) over the out-of-sample period, $h = 5$, 97.5% confidence. Source: author's estimation.

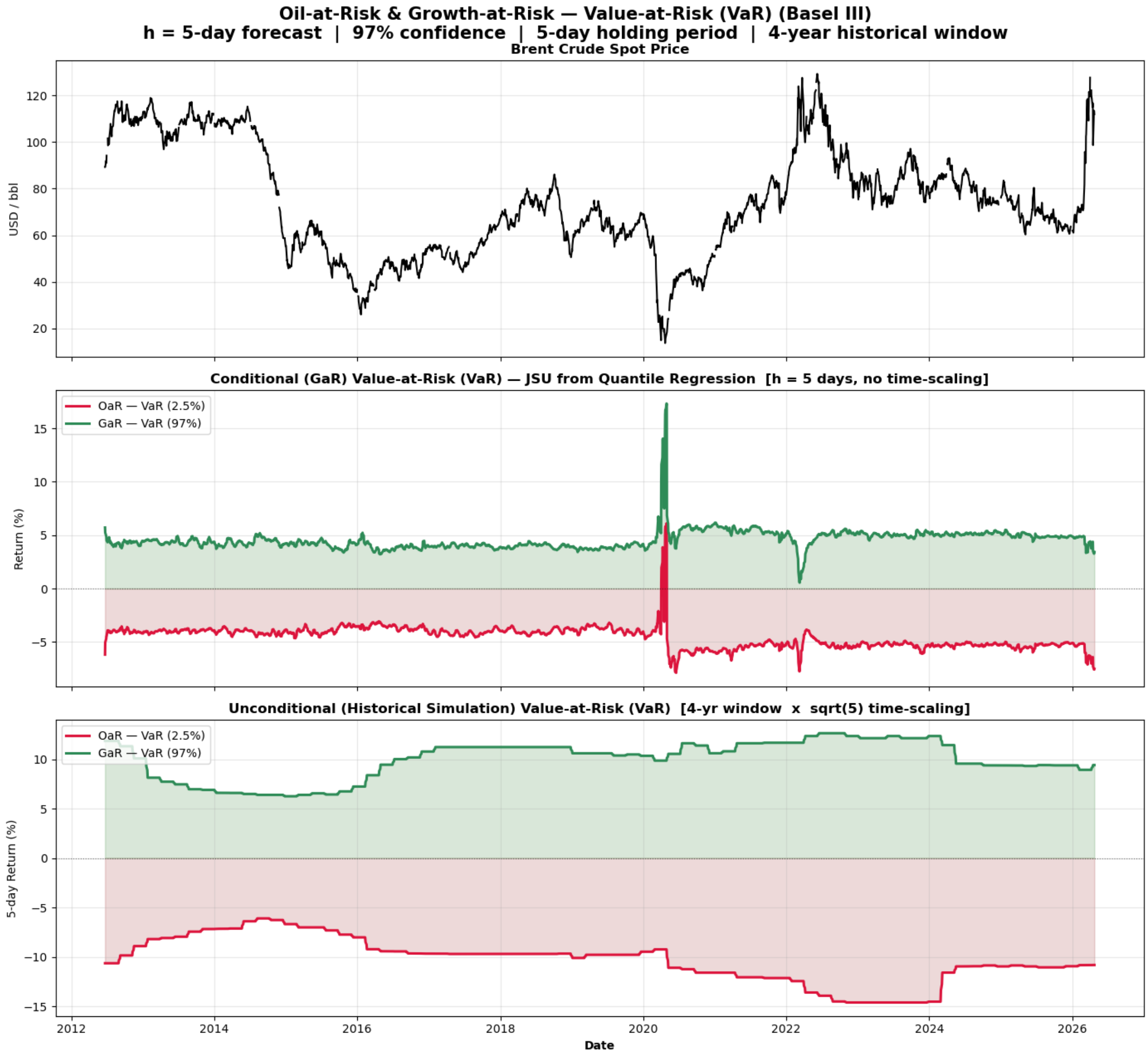

Figure 19: Conditional and unconditional Expected Shortfall (CVaR) over the out-of-sample period, $h = 5$, 97.5% confidence. Source: author's estimation.

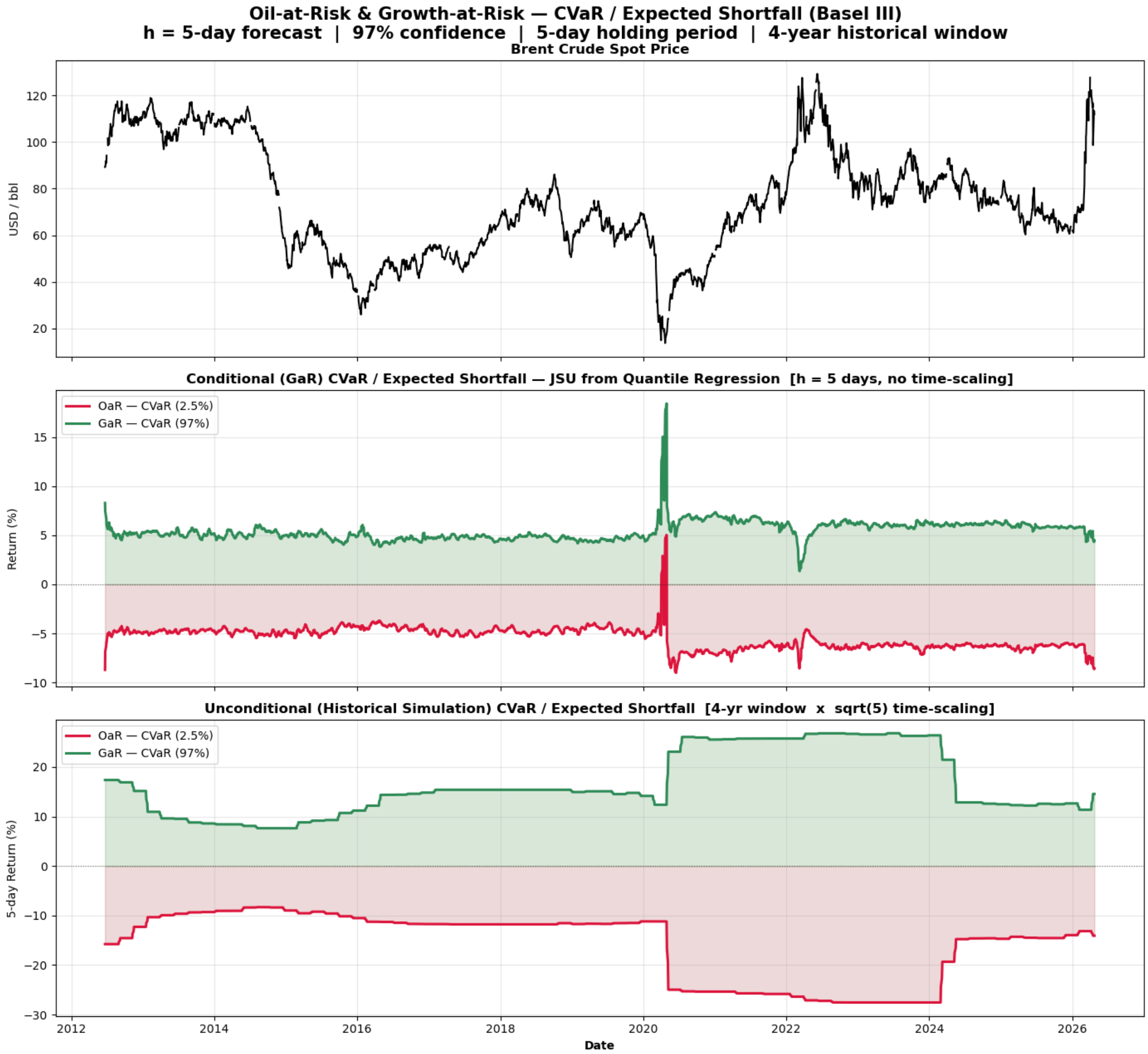

Figure 20: Cross-Correlation Function between the entropy indicators and Brent returns, with Bartlett 95% confidence bands. Source: author's estimation.